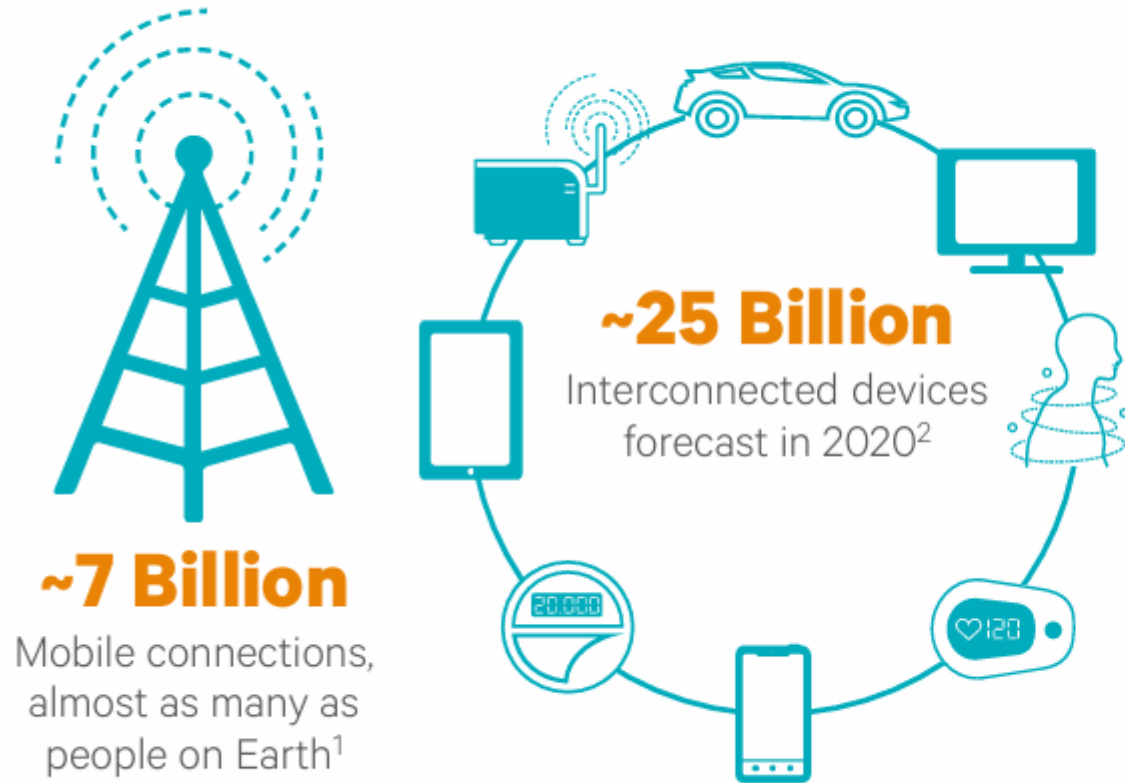


Chapter 4: Cellular Communications

Billions of Mobile Connections



Billions of Mobile Experiences



>100 Billion

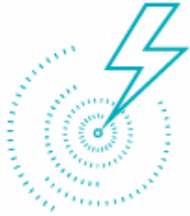
App downloads
completed in 2013³



~270 Billion

App downloads
expected in 2017³

The Mobile Experience



Mind-blowing Performance

with processing power greater than the most advanced super computers of the early 1990s¹



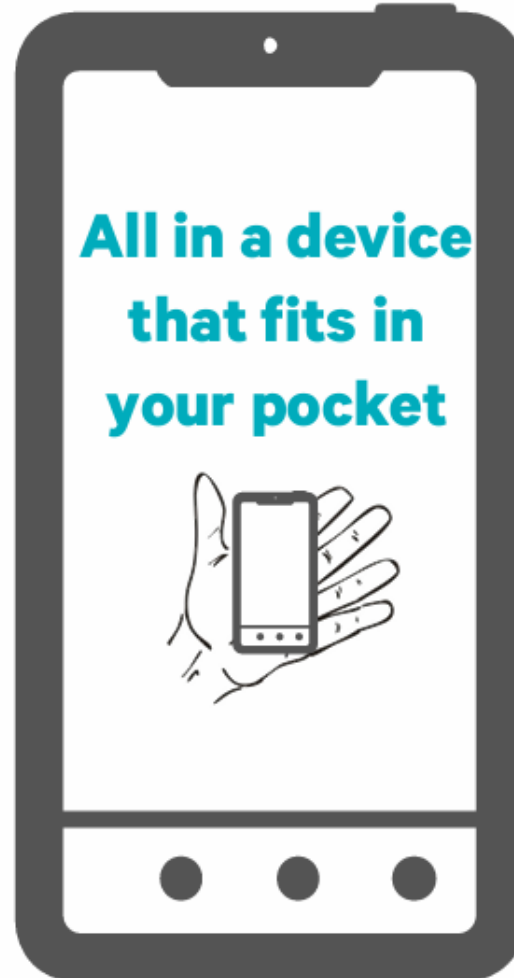
Jaw-dropping Graphics

with capability to process several thousand megapixels per second²



High Quality Multimedia²

4K UltraHD video player/recorder
HD gaming console
5.1/7.1 surround sound system
High resolution digital camera



Reliable Connectivity

overcoming signal loss resulting in receiving signal 100 trillion times weaker than when it originated³



Broadband Speeds

with blazing fast data rates capable of 300+ Mbps⁴

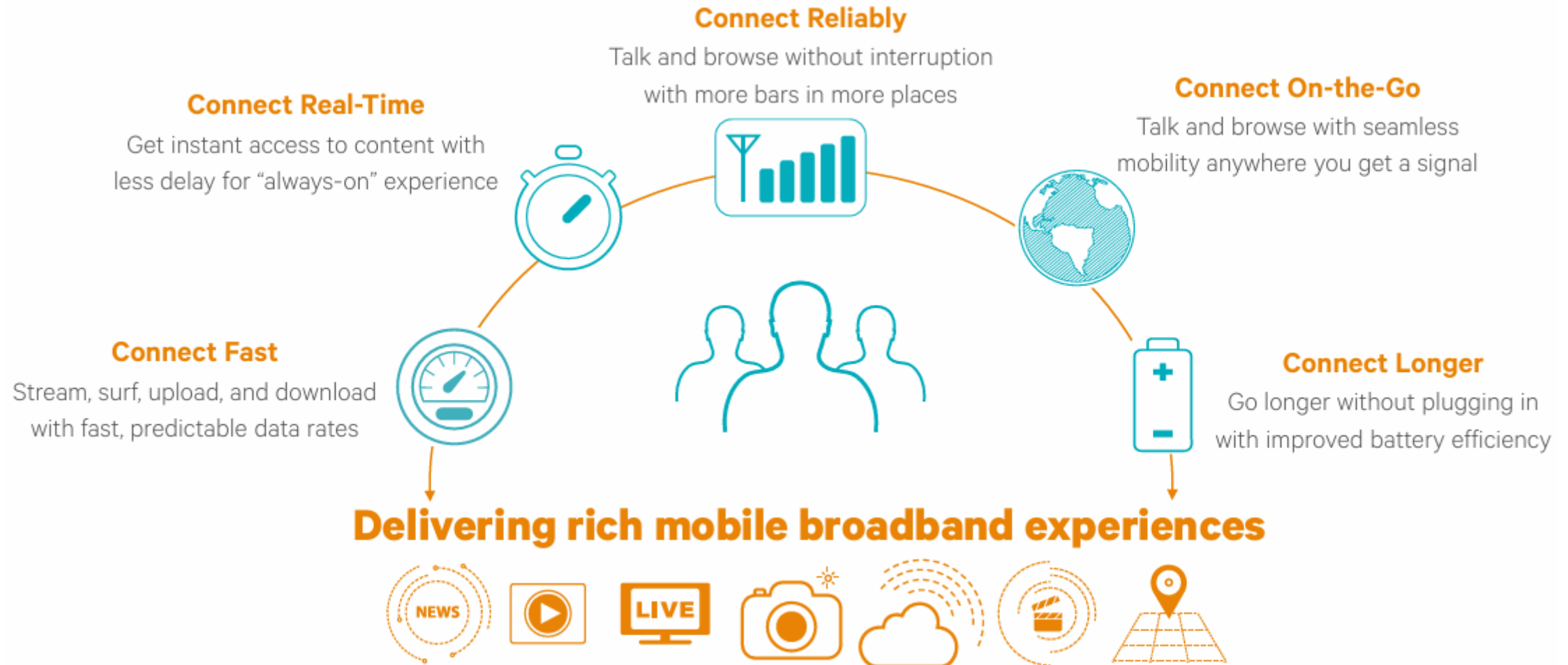


Long Battery Life

with ability to power all these amazing experiences with less energy than it takes to power a light bulb for 15 minutes⁵



Connectivity is the Foundation of a Great Mobile Experience



History of Wireless Communications



19th century

Radio telegraphy enables the transmission of data over radio waves. (Photo of Guglielmo Marconi, who sent the world's first radio message across open water on May 13, 1897)

1963
Institute of Electrical and Electronics Engineers (IEEE) forms

1971
ALOHAnet uses ultra-high frequency signals in the 400 MHz band to broadcast data packets

1991
Cellular shifts from analog first-generation cellular to 2G digital services

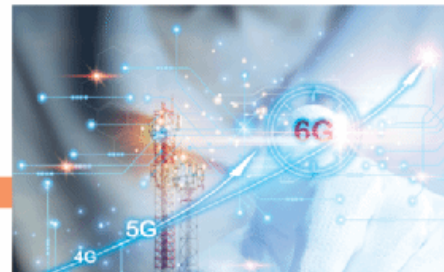
1998
3rd Generation Partnership Project (3GPP) forms to create specifications for 3G

Early 2000s
3G use increases with the popularity of smartphones

2002
WECA changes its name to Wi-Fi Alliance

2009
802.11n uses both the 2.4 GHz and 5 GHz bands and introduces multiple input, multiple output (MIMO)

2013
802.11ac brings speeds up to 1 Gbps and introduces multi-user MIMO



1985
FCC opens the 900 MHz, 2.4 GHz and 5 GHz bands

1997
IEEE codifies the first 802.11 wireless standard, which supports speeds of 2 Mbps

1999
— 802.11a offers up to 54 Mbps in the 5 GHz band
— 802.11b offers 11 Mbps in the 2.4 GHz band
— Vendors form Wireless Ethernet Compatibility Alliance (WECA)

2003
802.11g introduces speeds of 22 Mbps in 2.4 GHz band

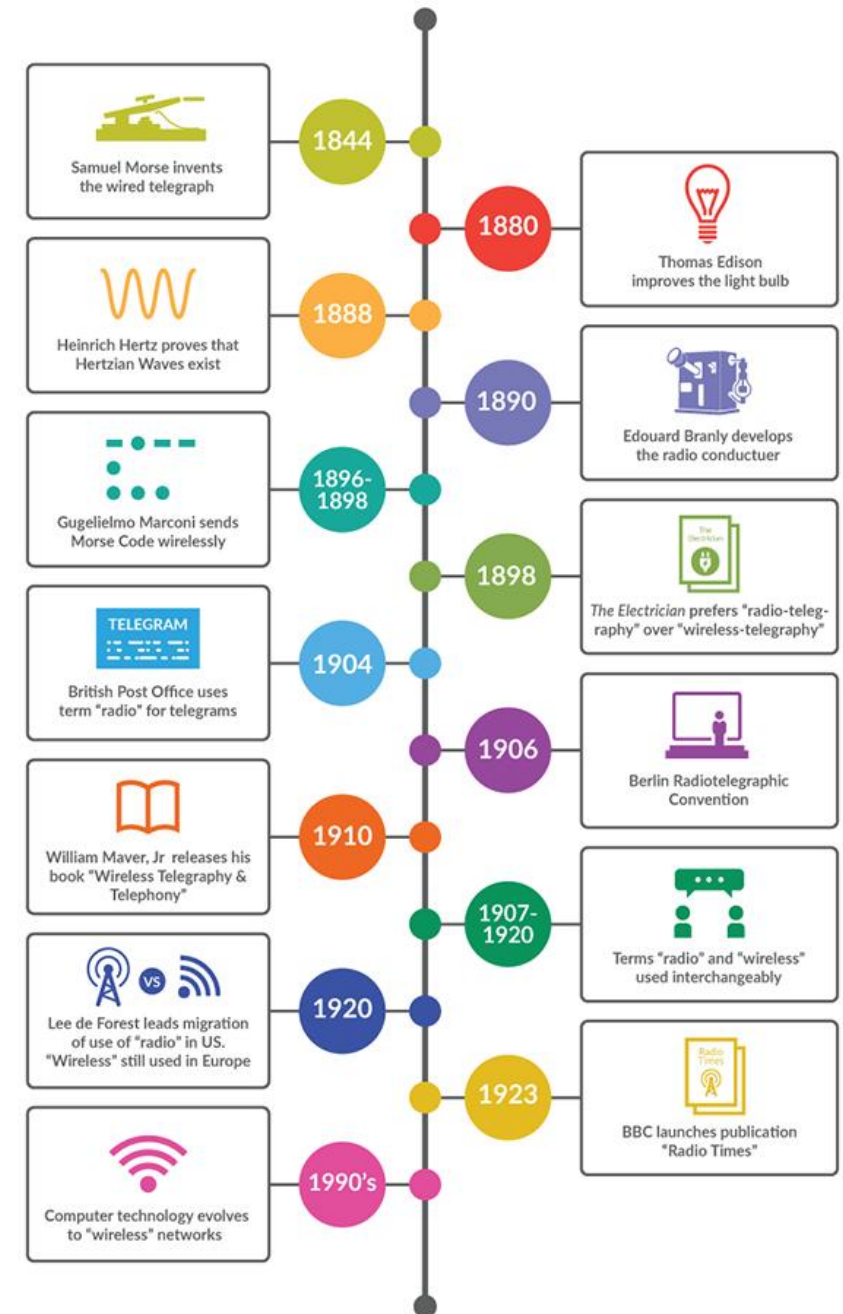
2009 to early 2010s
4G and 4G LTE offer high cellular speeds and grow more popular

2017
3GPP approves its Release 15 package, which introduces 5G

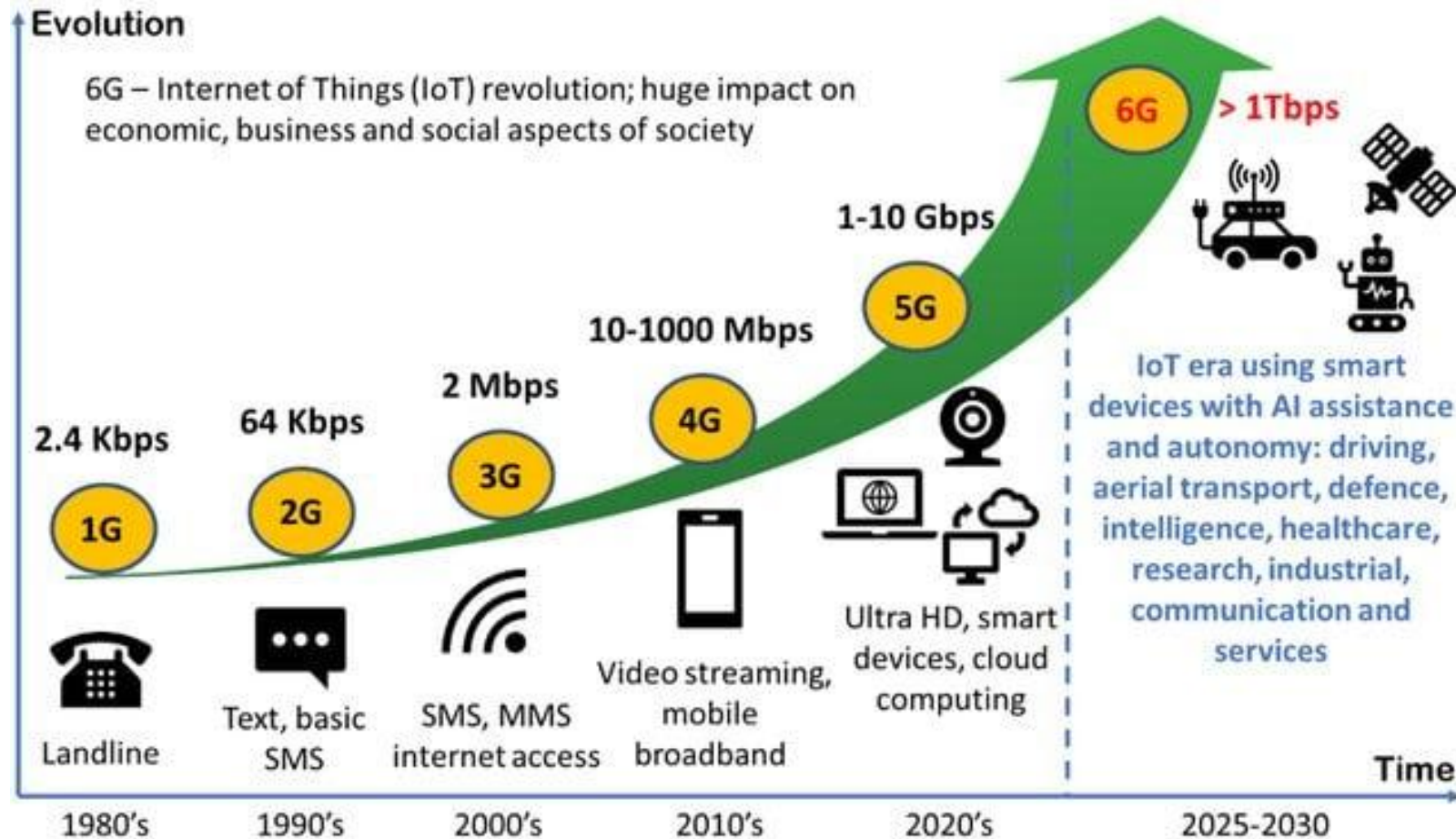
2019
802.11ax boosts efficiency and supports more devices

2024
IEEE expects to release 802.11be, which will operate in 2.4 GHz, 5 GHz and 6 GHz bands

2030
6G expected to launch



Trends in Wireless Communications



Factors Affecting Wireless System Design

Frequency allocations

- What range to operate?
- Need licenses.

Multiple access mechanism

- How do users share the medium without interfering?

Antennas and propagation

- What distances?
- Possible channel errors introduced.

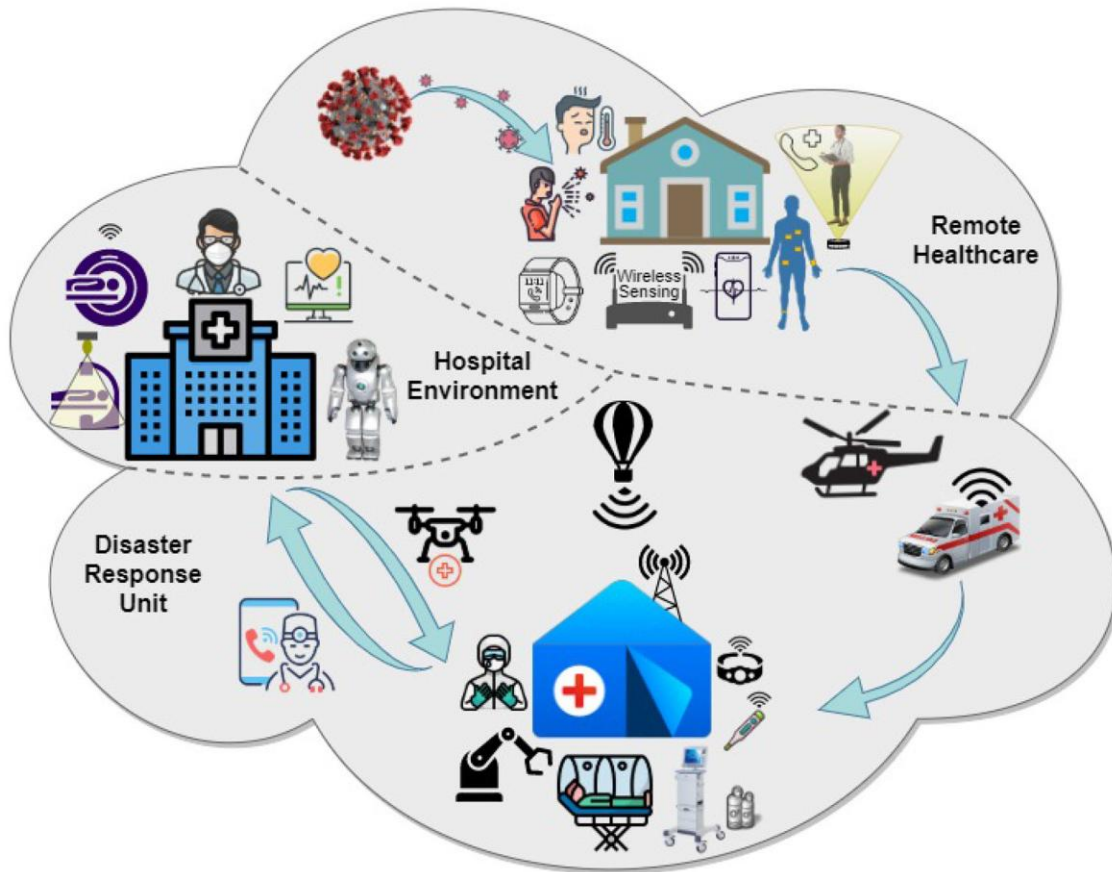
Signals encoding

- How to improve the data rate?

Error correction

- How to ensure that bandwidth is not wasted?

Issues Vital to Cellular



Frequency allocation

- Licensed
- Many providers

Multiple Access

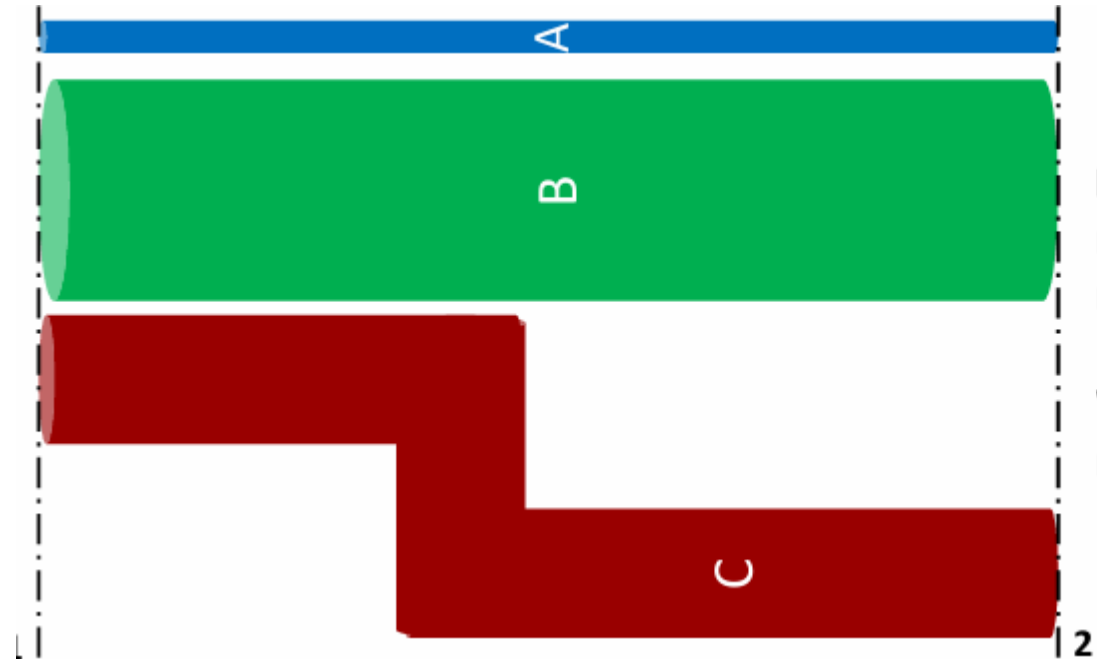
- Many users
- Wide area of coverage
- Traffic management

Location management

- High mobility (in cars, trains)
- Multiple suppliers
- Handoff management, roaming

Latency & Jitter

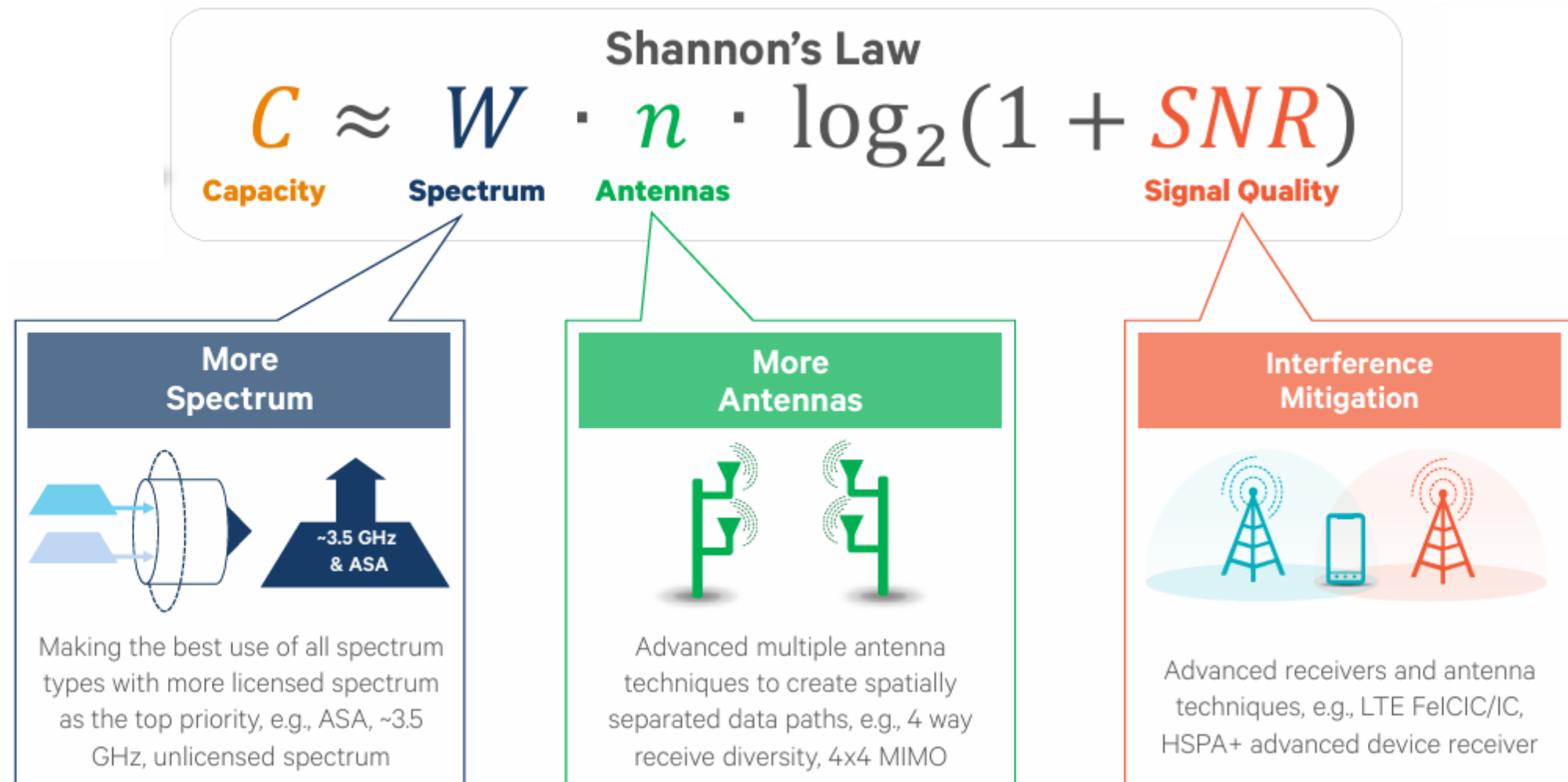
- **Latency** is generally defined as the time it takes for a source to send a packet of data to a receiver. In simple terms, half of Ping time.
 - ✓ This is also referred to as one way latency. Sometimes the term **Round trip latency or round trip time (RTT)** is also used to define latency.
 - ✓ This is the same as ping time.
- **Jitter** is defined as the variation in the delay (or latency) of received packets. It is also referred to as 'delay jitter'.



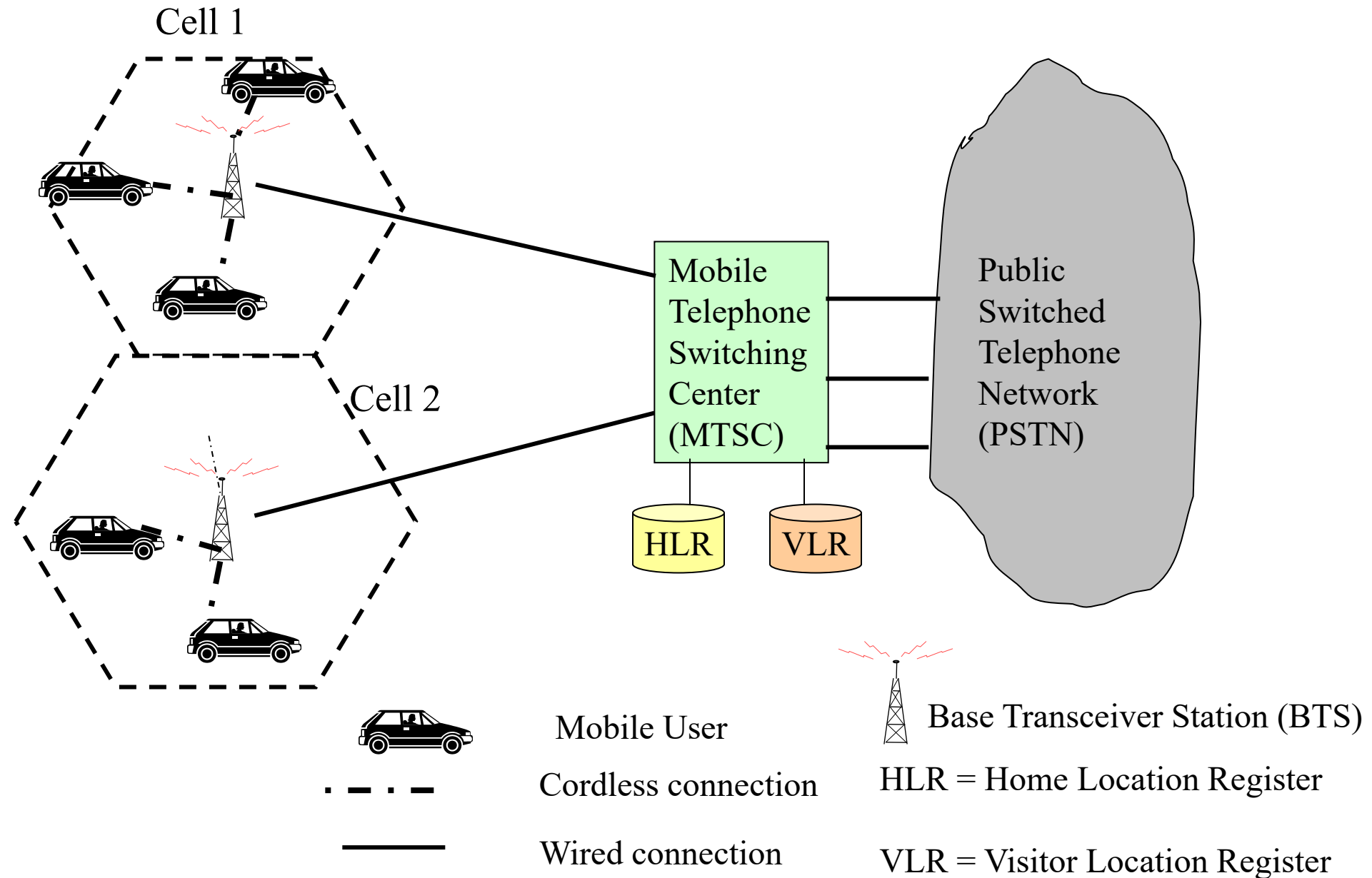
Bandwidth is often referred to as a measure of capacity.

While **Latency** is a measure of delay.

Data Capacity



Cellular Network Concept



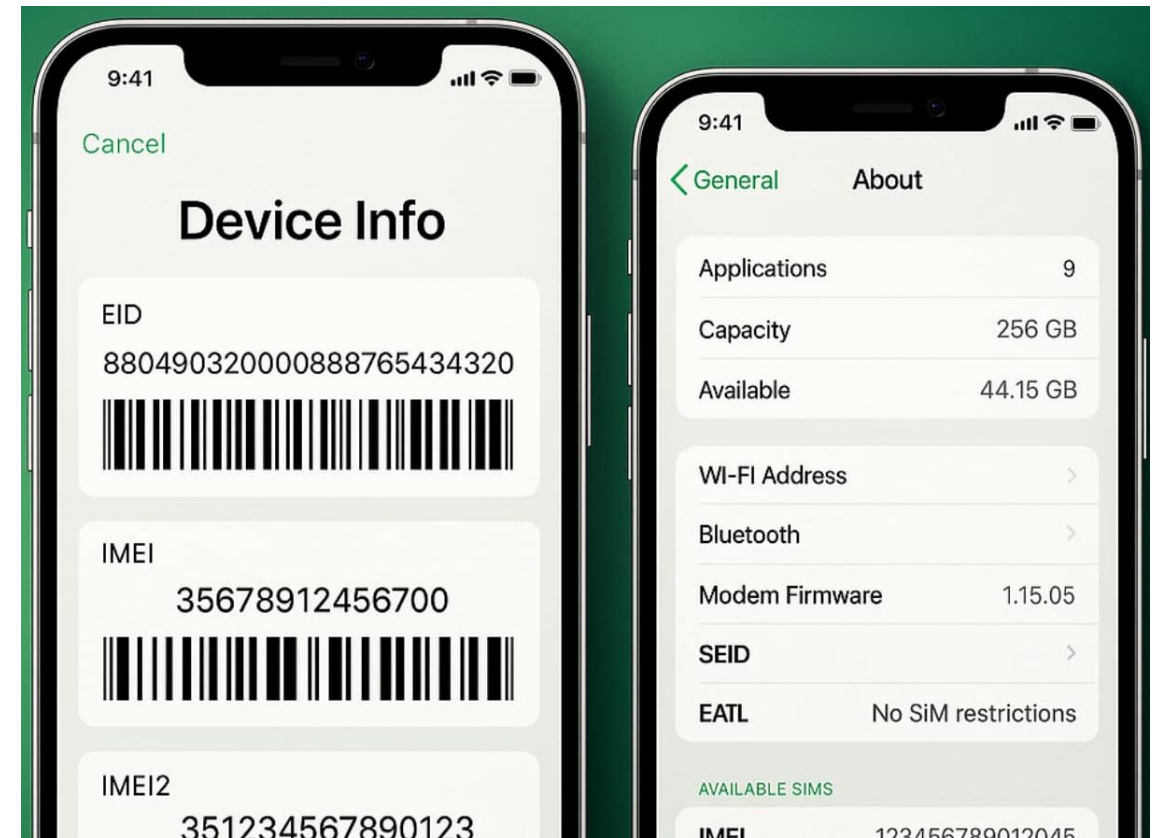
Addresses and Identifiers

International Mobile Station Equipment Identity (IMEI)

- A serial number
 - distinctively identifies a mobile station internationally
 - 15 digit no.
- Allocated by the equipment manufacturer
- Registered by the network operator,

Parts of IMEI

- **Type Approval Code (TAC)** – 8 decimal places, centrally assigned.
 - 1st two digits indicate the specific GSMA-approved group that authorized the device. Remaining six identify the manufacturer and the specific model of the phone.
- **Serial Number (SNR)** – 6 decimal places, assigned by the manufacturer.
- **Check Digit (CD)** – 1 decimal place.
 - checksum calculated using the Luhn algorithm.



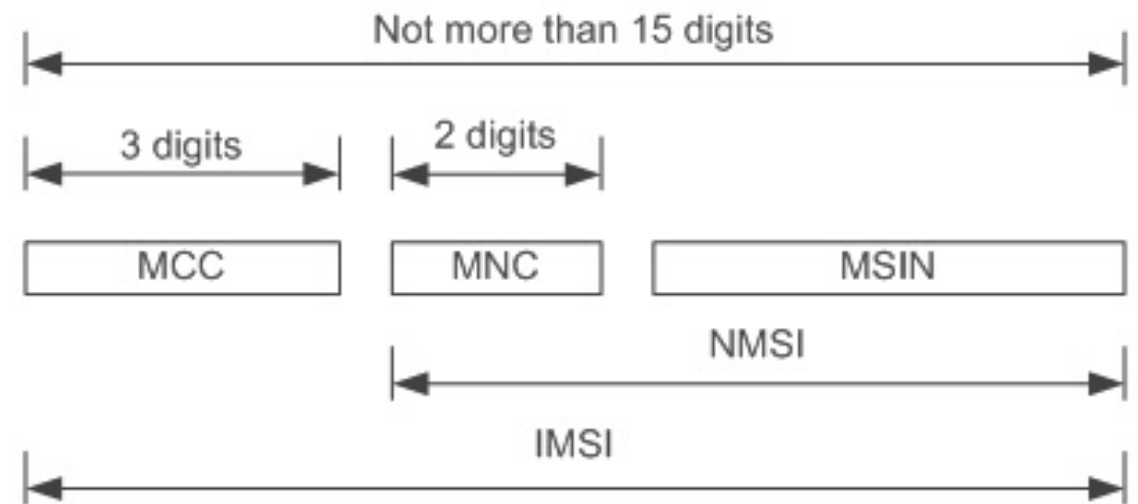
Addresses and Identifiers

International Mobile Subscriber Identity (IMSI)

- Unique sim card number
- Stored in their Subscriber Identity Module (SIM)

Parts of IMSI

- **Mobile Country Code (MCC)** – 3 decimal places, internationally standardized.
- **Mobile Network Code (MNC)** – 2 decimal places, for unique identification of mobile network within the country.
- **Mobile Subscriber Identification Number (MSIN)** – Maximum 10 decimal places, identification number of the subscriber in the home mobile network



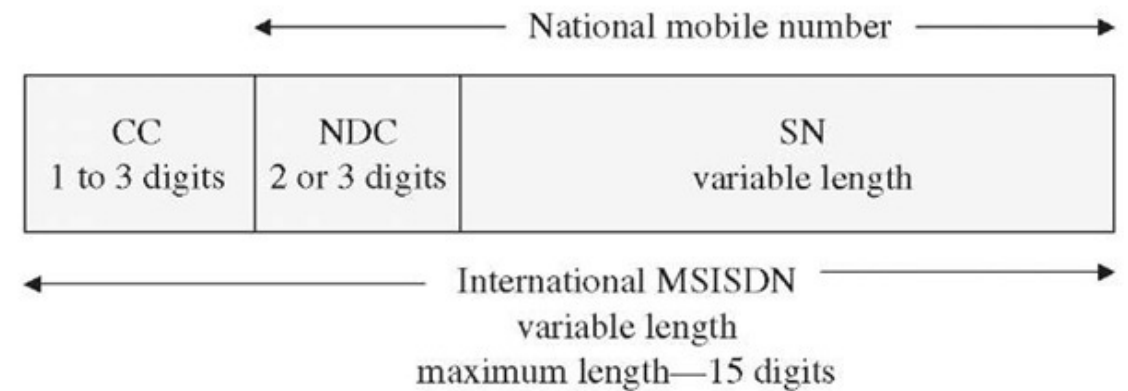
Addresses and Identifiers

Mobile Subscriber ISDN Number (MSISDN)

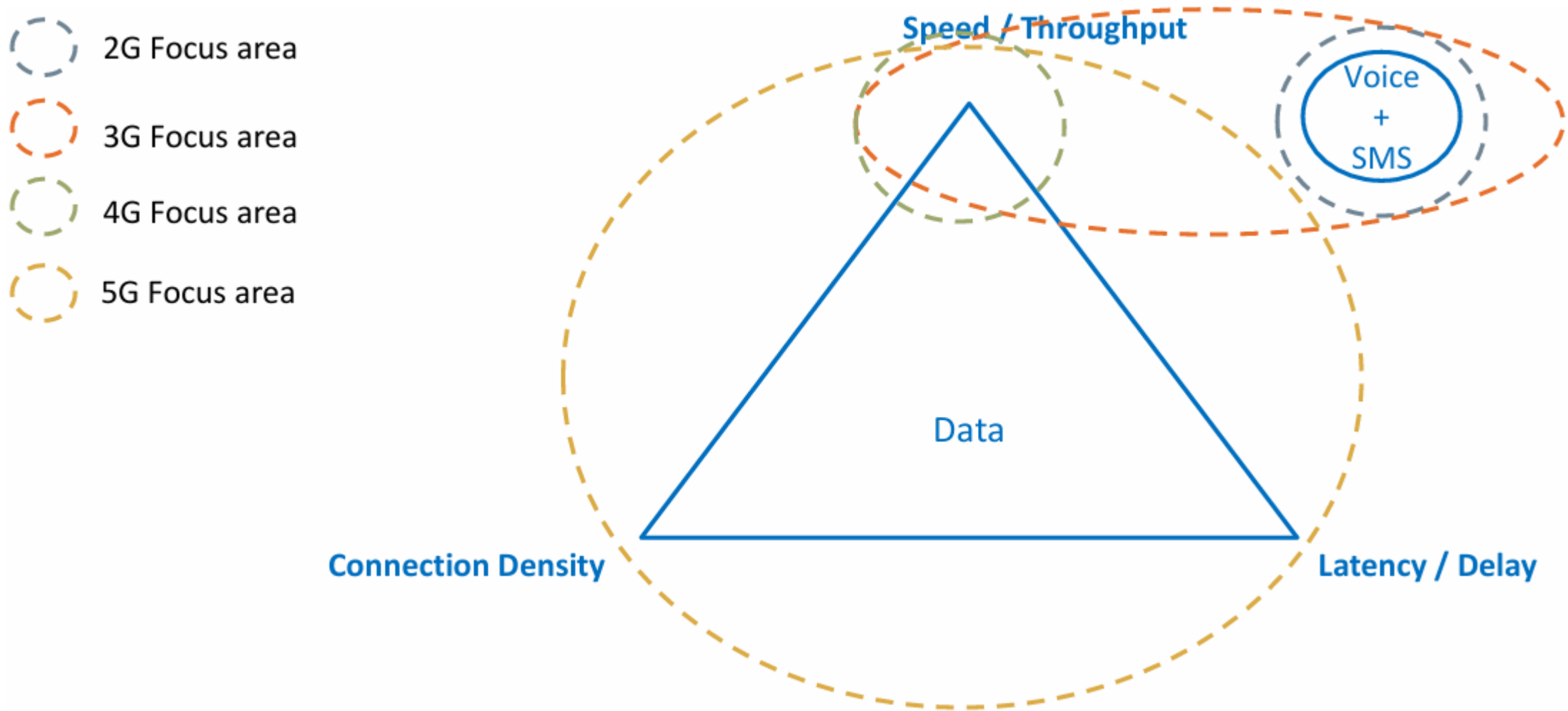
- Your Phone number
- The authentic telephone number of a mobile station is the Mobile Subscriber ISDN Number (MSISDN).

Parts of IMSI

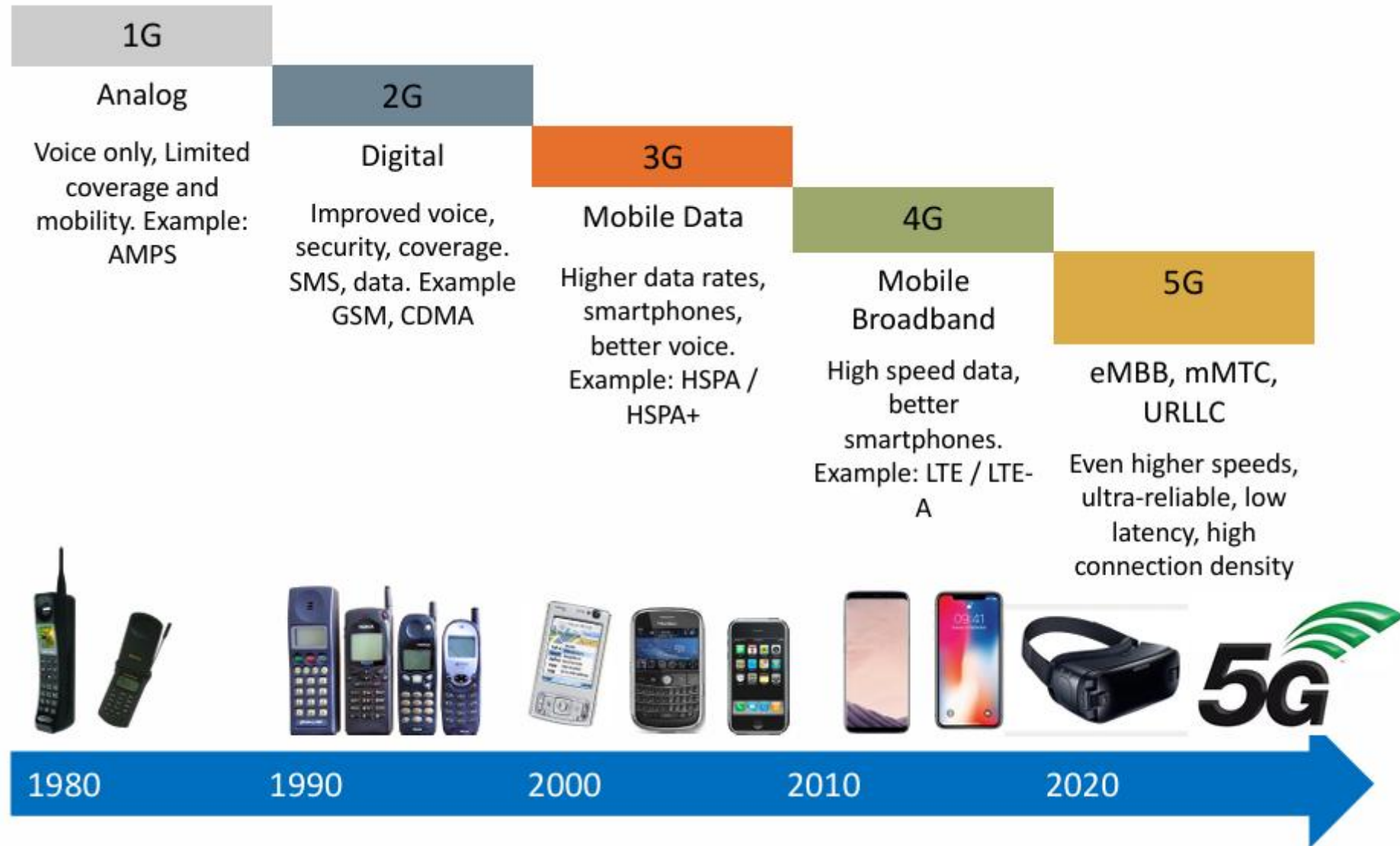
- **Country Code (CC)** – Up to 3 decimal places.
- **National Destination Code (NDC)** – Typically 2-3 decimal places.
- **Subscriber Number (SN)** – Maximum 10 decimal places.



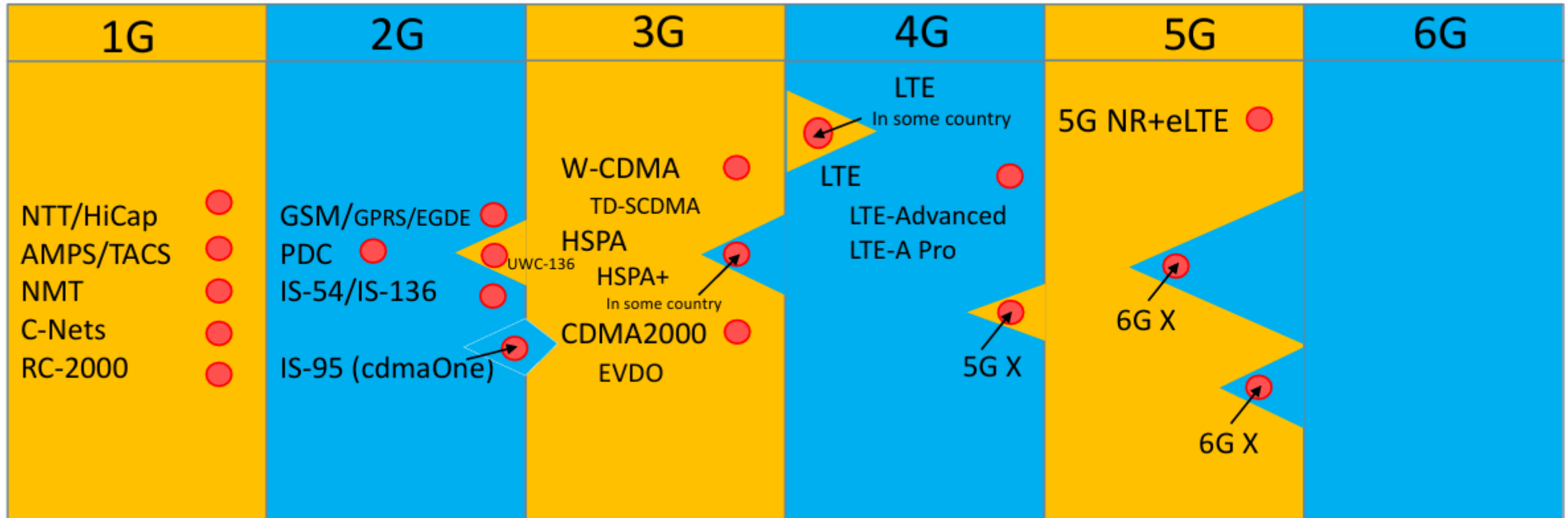
Focus area for Different Technology Generations



Mobile Technology Evolution



Mobile Generation by Standards



- There is constant battle between standards, technology and marketing with regards to naming for different generations. Marketing generally wins!

1G-5G Network Architectures

First Generation (1G)

- Mobile 1G established the foundation of mobile



- 1G analog voice was amazing, but limited

1

Licensed Spectrum

Cleared spectrum for exclusive use by mobile technologies



Operator-deployed **base stations** provide access for subscribers

2

Frequency Reuse

Reusing frequencies without interference through geographical separation

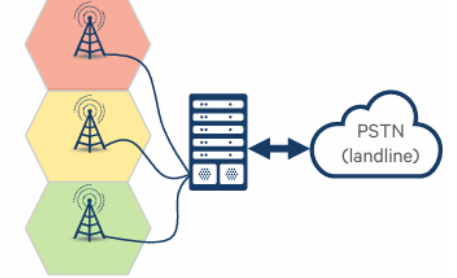


Neighboring **cells** operate on different frequencies to avoid interference

3

Mobile Network

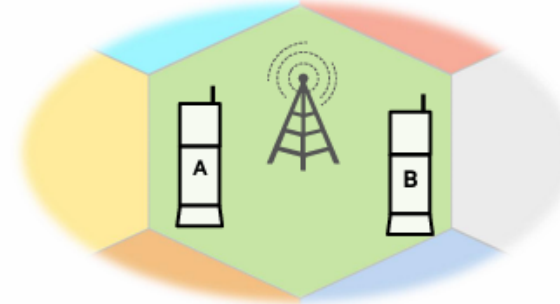
Coordinated network for seamless access and seamless mobility



Integrated, transparent **backhaul network** provides seamless access

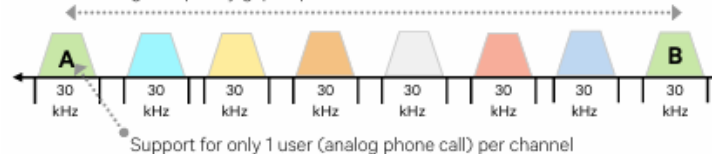
Limited Capacity

Analog transmissions are inefficient at using limited spectrum



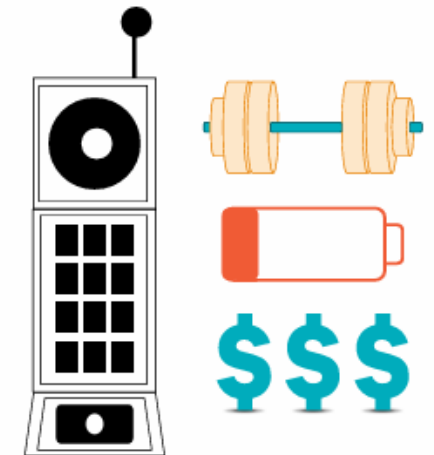
Frequency Division Multiple Access (FDMA)*

Large frequency gap required between users to avoid interference

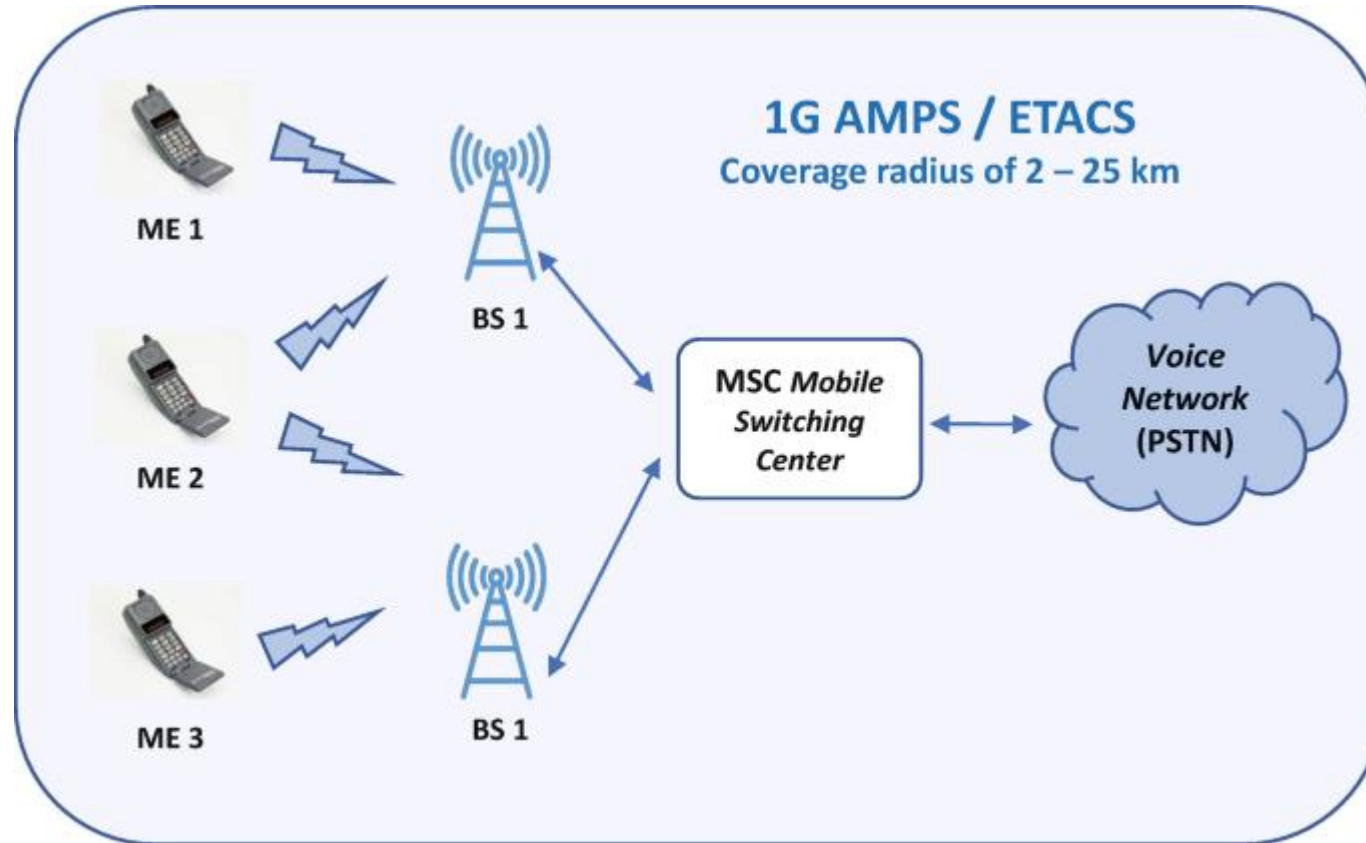


Limited Scalability

Analog devices are large/heavy, power inefficient, and high cost



1G Network Architecture

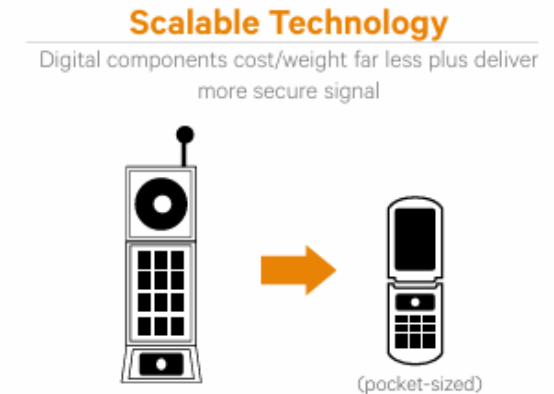
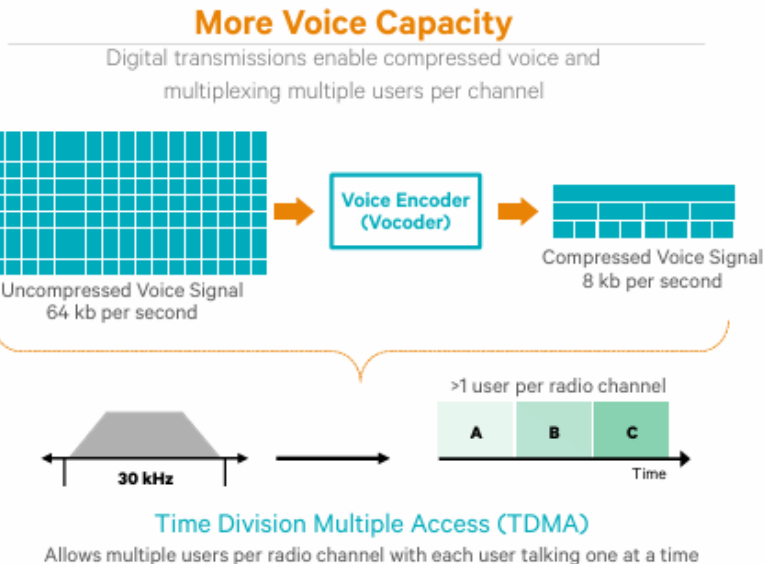
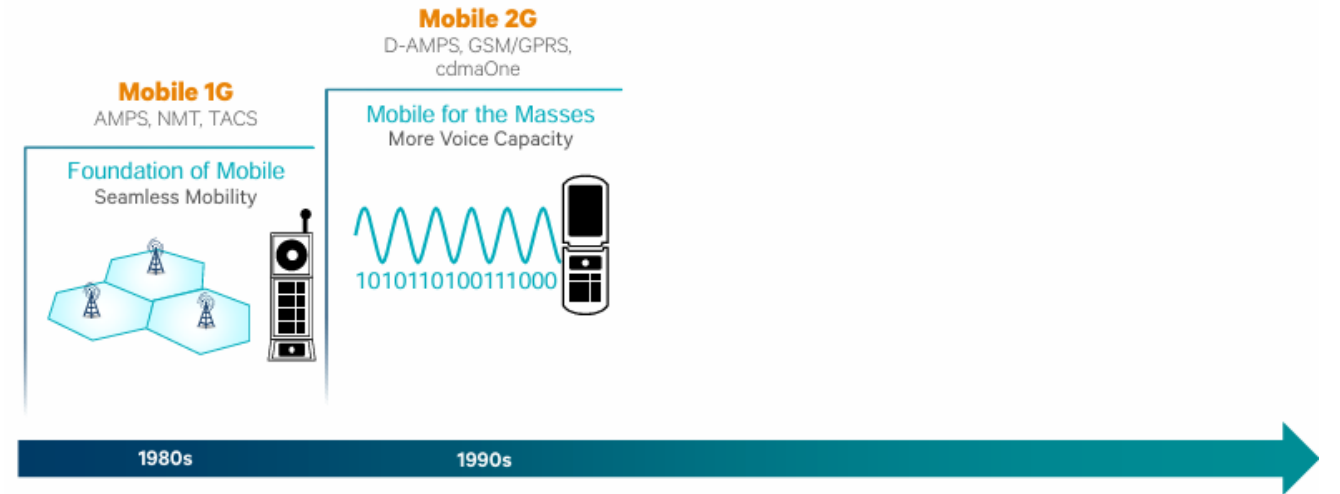


Second Generation (2G)

- Mobile 2G digital technologies increased voice capacity
 - ✓ Delivering mobile voice services to the masses – more people, in more places

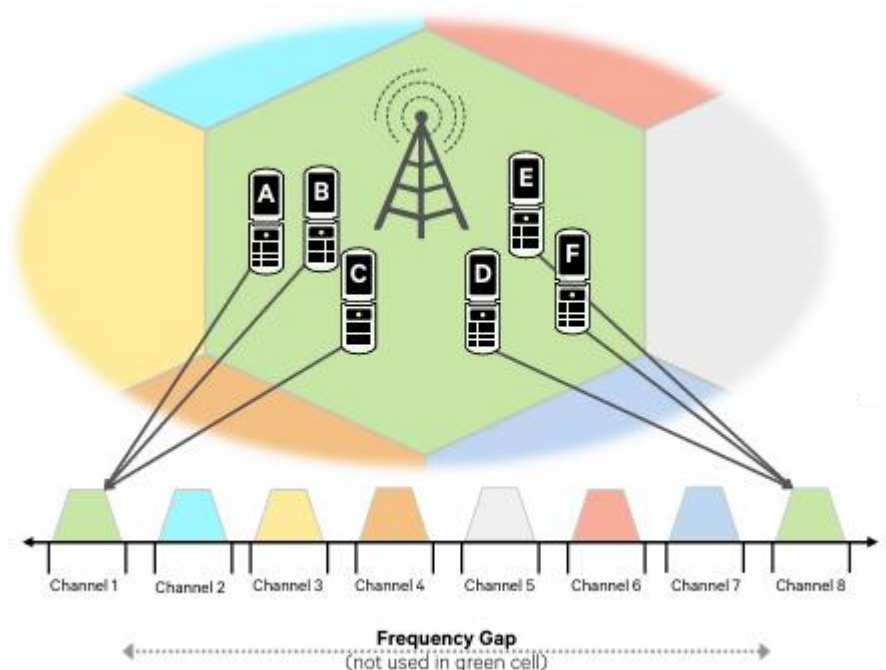


- Mobile 2G digital wireless technologies enabled more users
 - ✓ Initial 2G technologies (D-AMPS, GSM) based on TDMA



Second Generation (2G)

- Different Mobile 2G TDMA techniques were standardized



- TDMA still required large frequency gaps to reduce interference

Mobile 1G (Analog)

AMPS, NMT, TACS



Mobile 2G (Digital)

D-AMPS

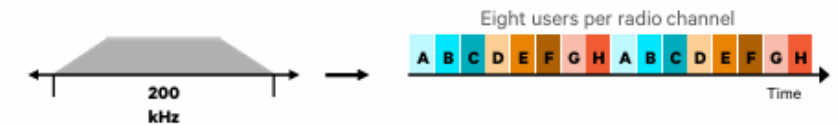
Standardized as IS-54 by TIA in 1992
Mainly in North America
No longer utilized



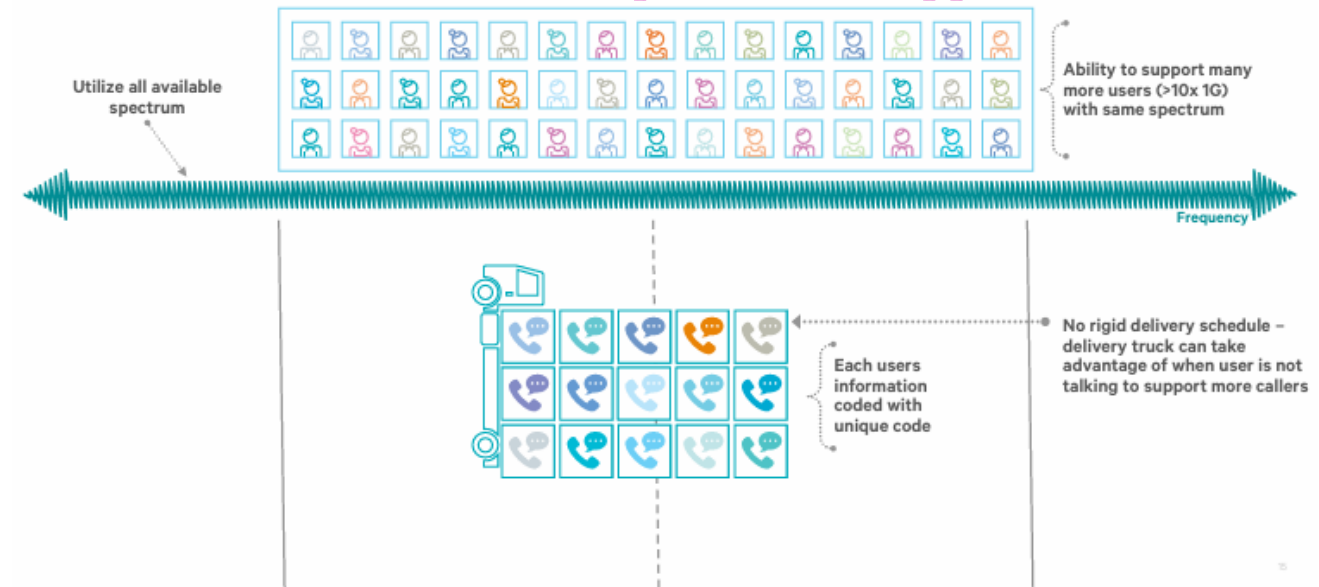
Mobile 2G (Digital)

GSM

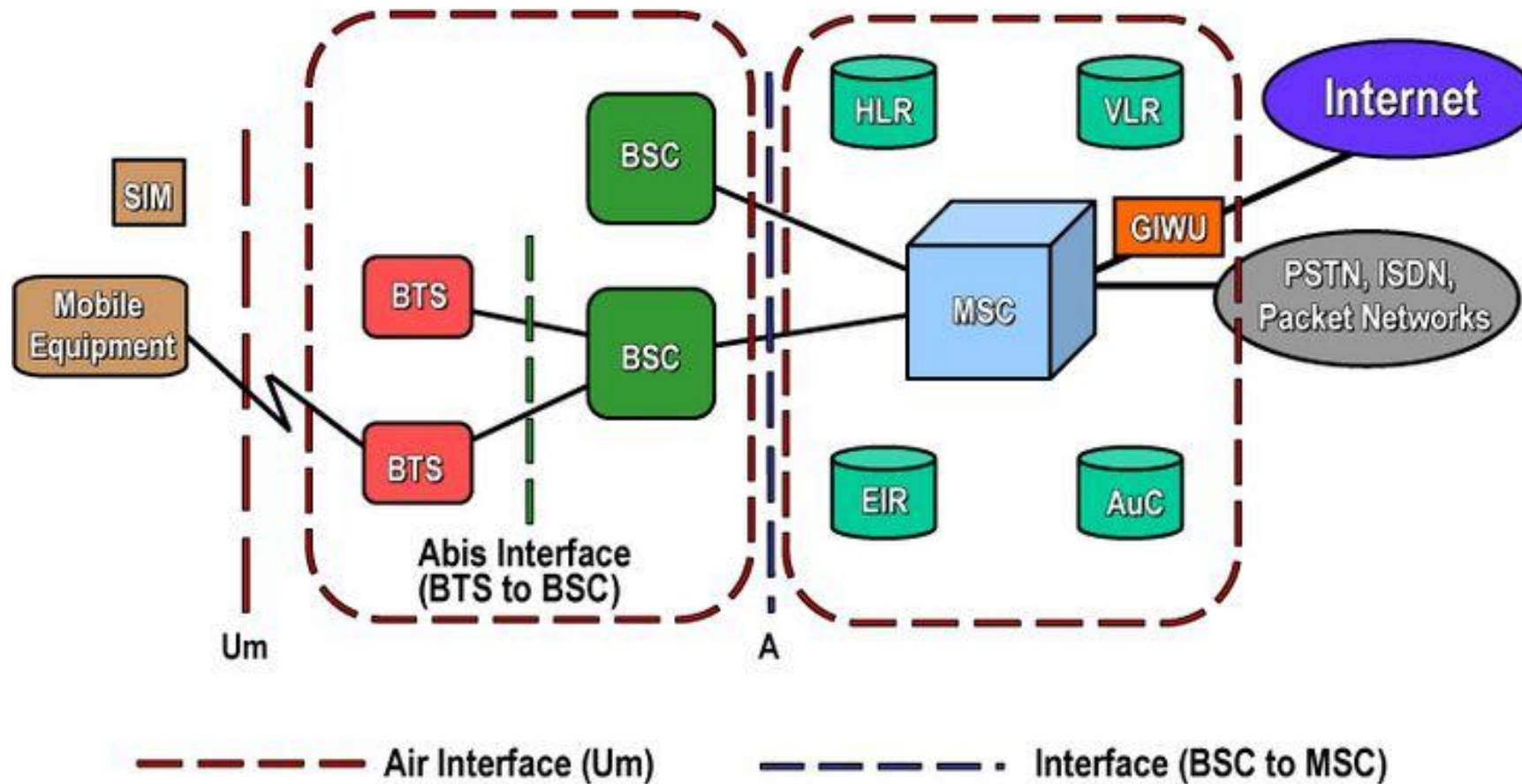
Standardized by ETSI in 1990 (phase 1)
Initiated in Europe
Still widely used today (>4B connections WW)
Simple data services with GPRS



- CDMA utilizes all the available spectrum to support more users



2G (GSM) Network Architecture

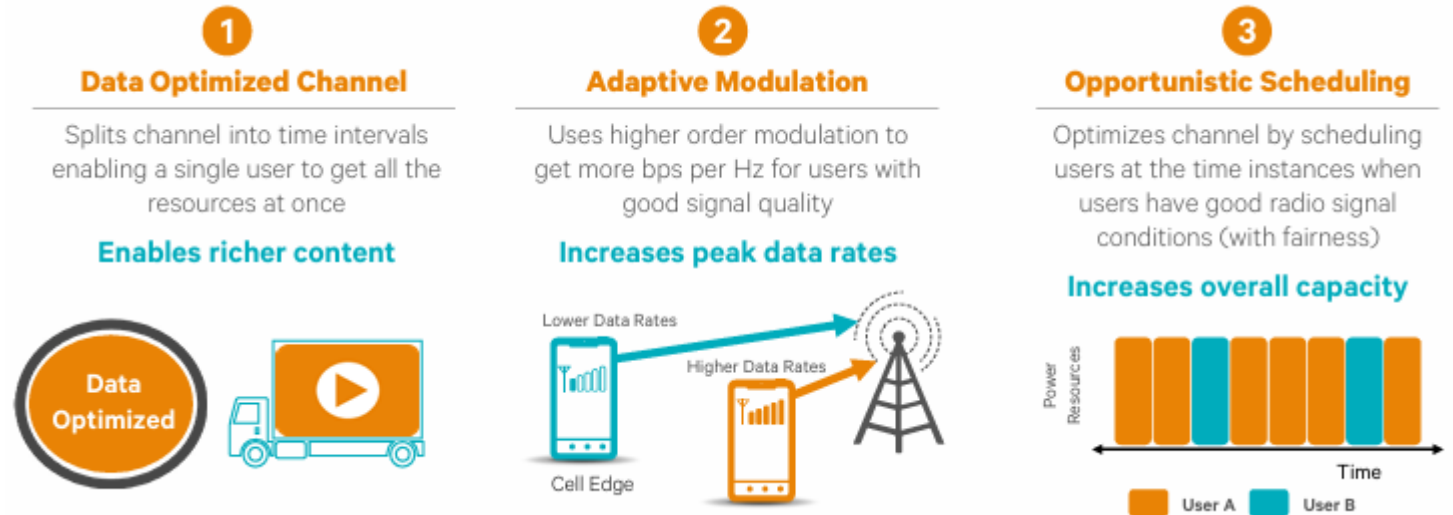
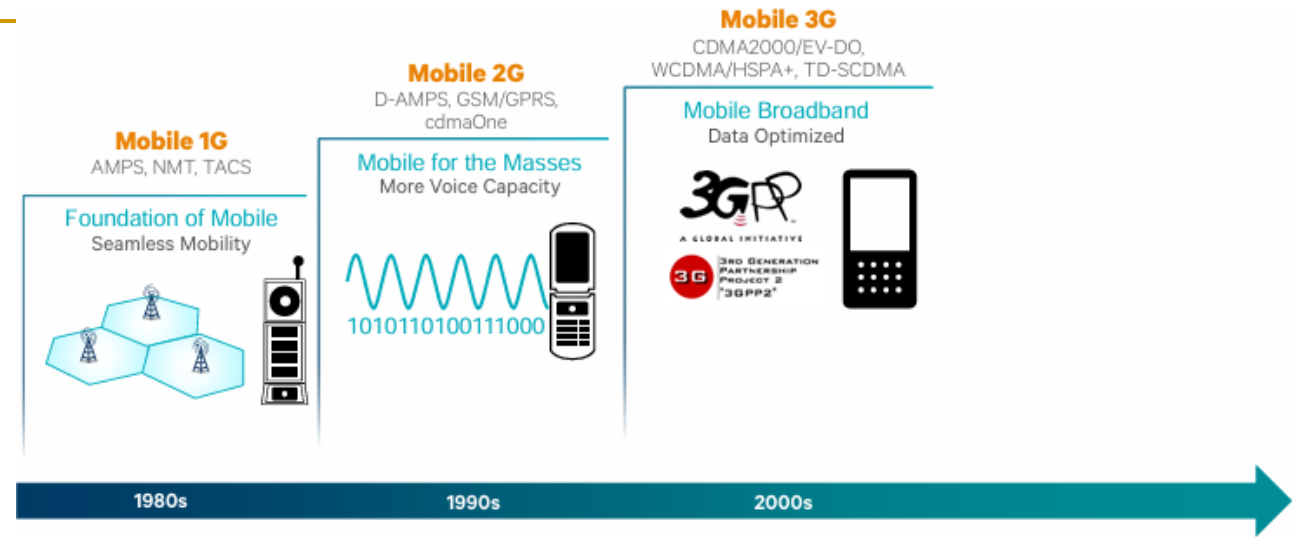


Third Generation (3G)

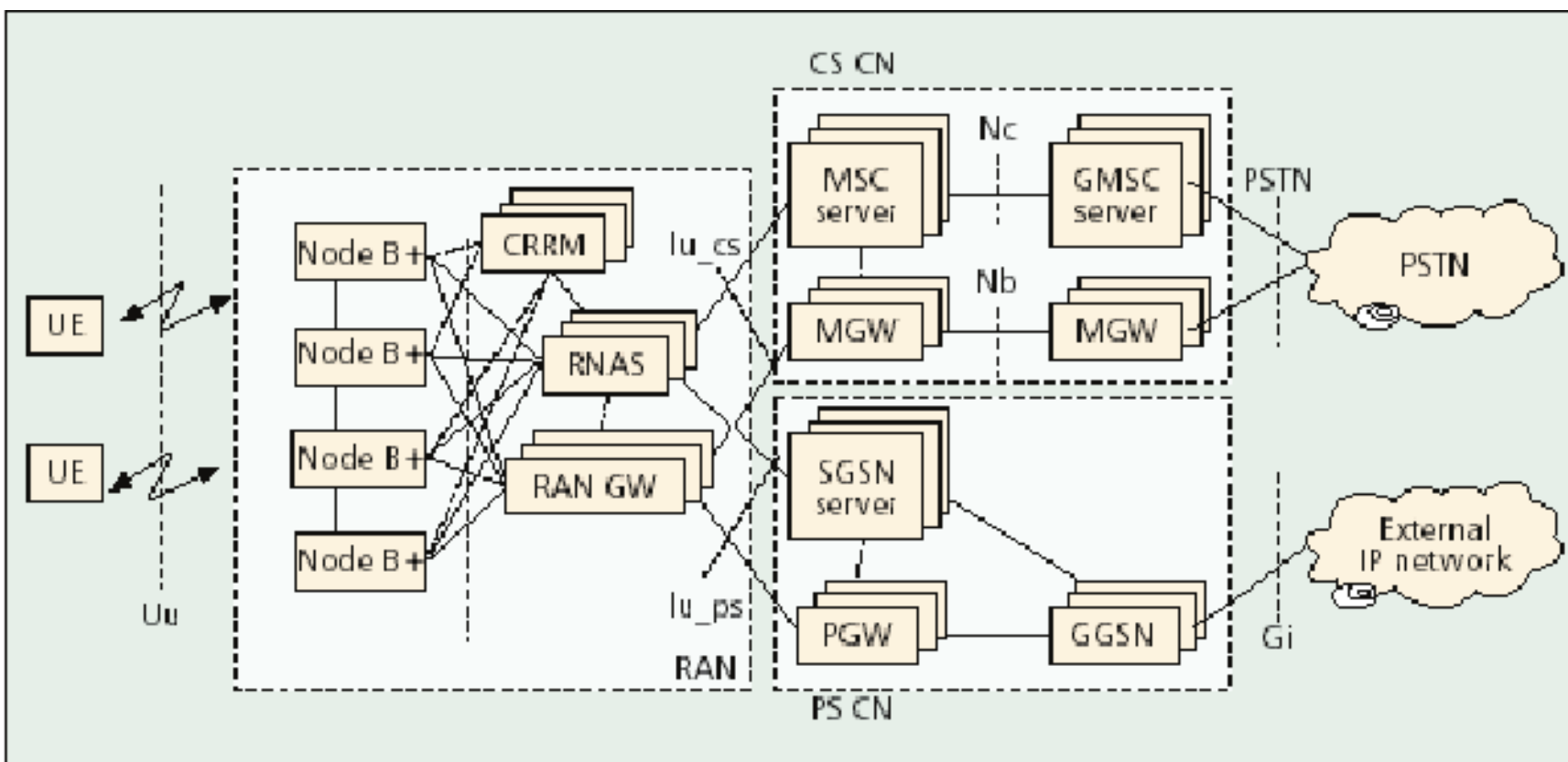
- Mobile 3G evolved mobile for data
 - ✓ Introducing high-speed internet access for the first time



- EV-DO inventions are the foundation to mobile broadband



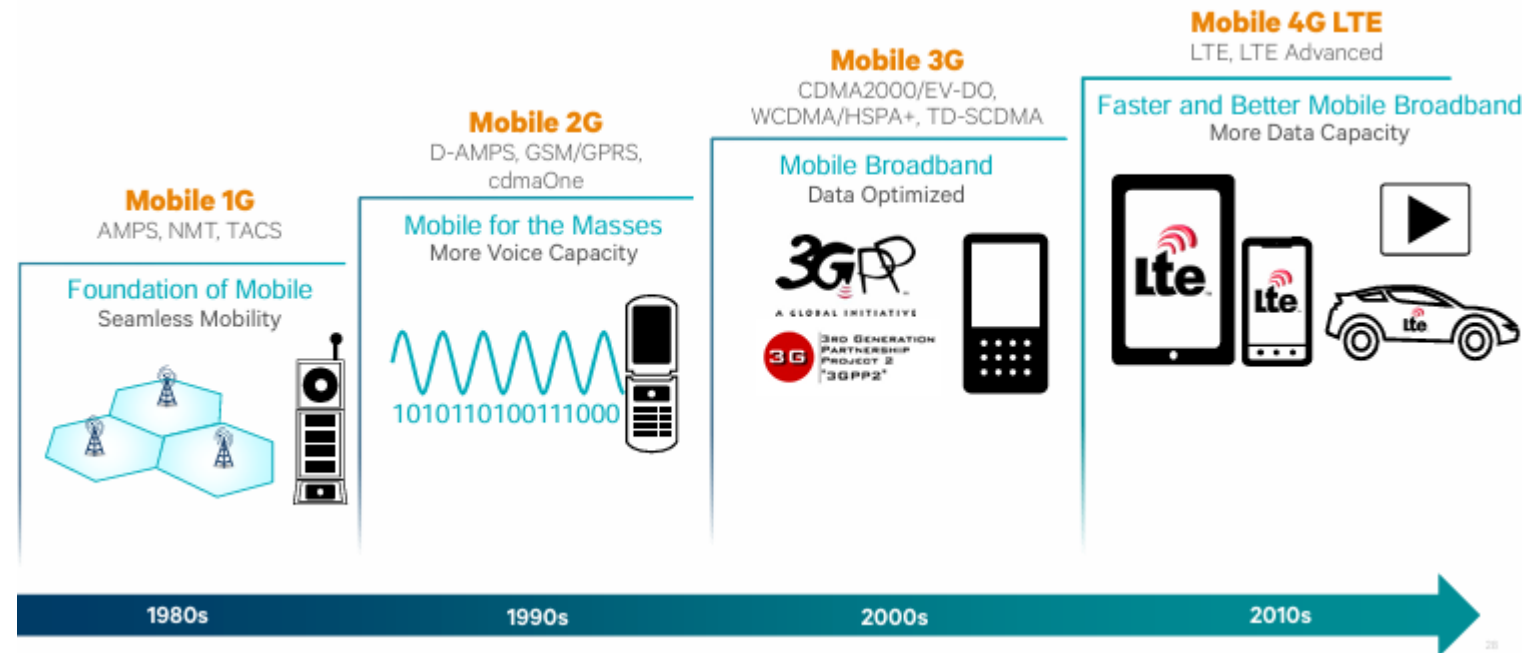
3G Network Architecture



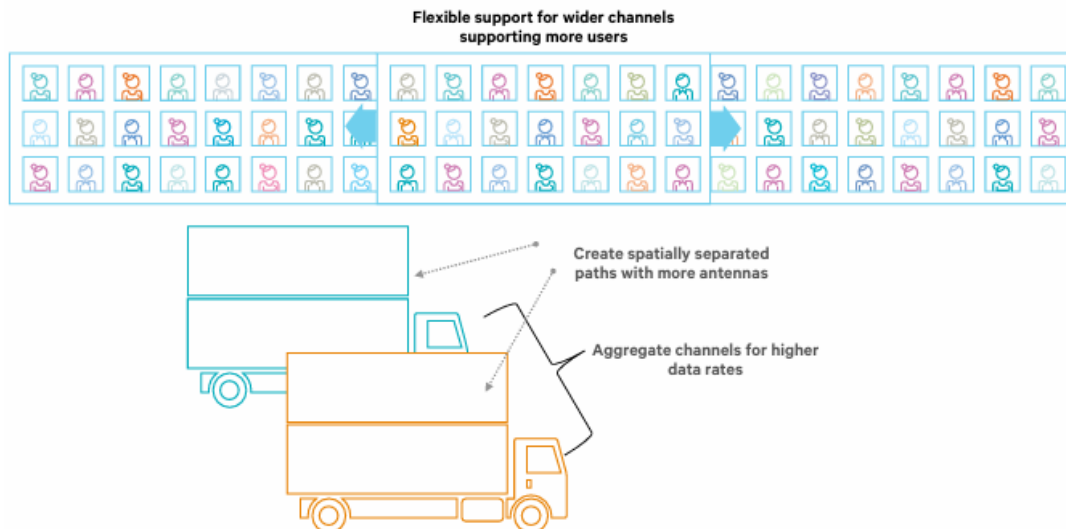
Node B	Base Station
CRRM	Common Radio Resource Management
RNAS	Radio Network Access Server
RAN GW	Radio Access Network Gateway
Iu_cs	Circuit-Switched Interface
Iu_ps	Packet-Switched Interface
CS CN	Circuit-Switched Core Network
MSC Server	Mobile Switching Center Server
MGW	Media Gateway
GMSC Server	Gateway Mobile Switching Center Server
Nc	Network Control Interface
Nb	Bearer Control Interface
PSTN	Public Switched Telephone Network
PS CN	Packet-Switched Core Network
SGSN Server	Serving GPRS Support Node
PGW	Packet Gateway
GGSN	Gateway GPRS Support Node
Gi	Gateway Interface
External IP Network	Internet / IP Network

Fourth Generation (4G)

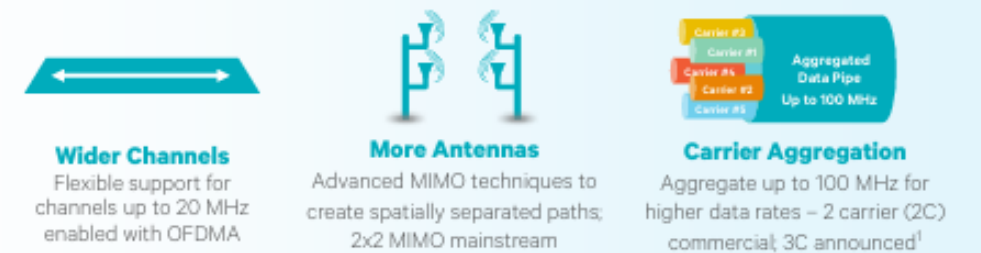
- Mobile 4G LTE is evolving to provide more data capacity
 - ✓ Delivering faster and better mobile broadband experiences



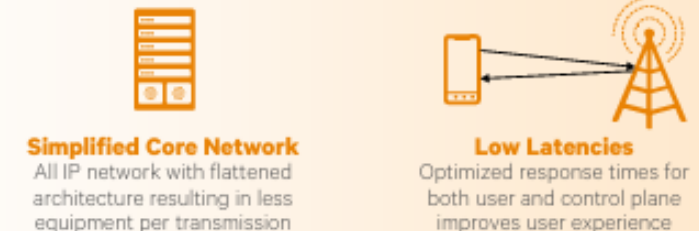
- Mobile 4G LTE delivers more data capacity



**Connect
Faster**

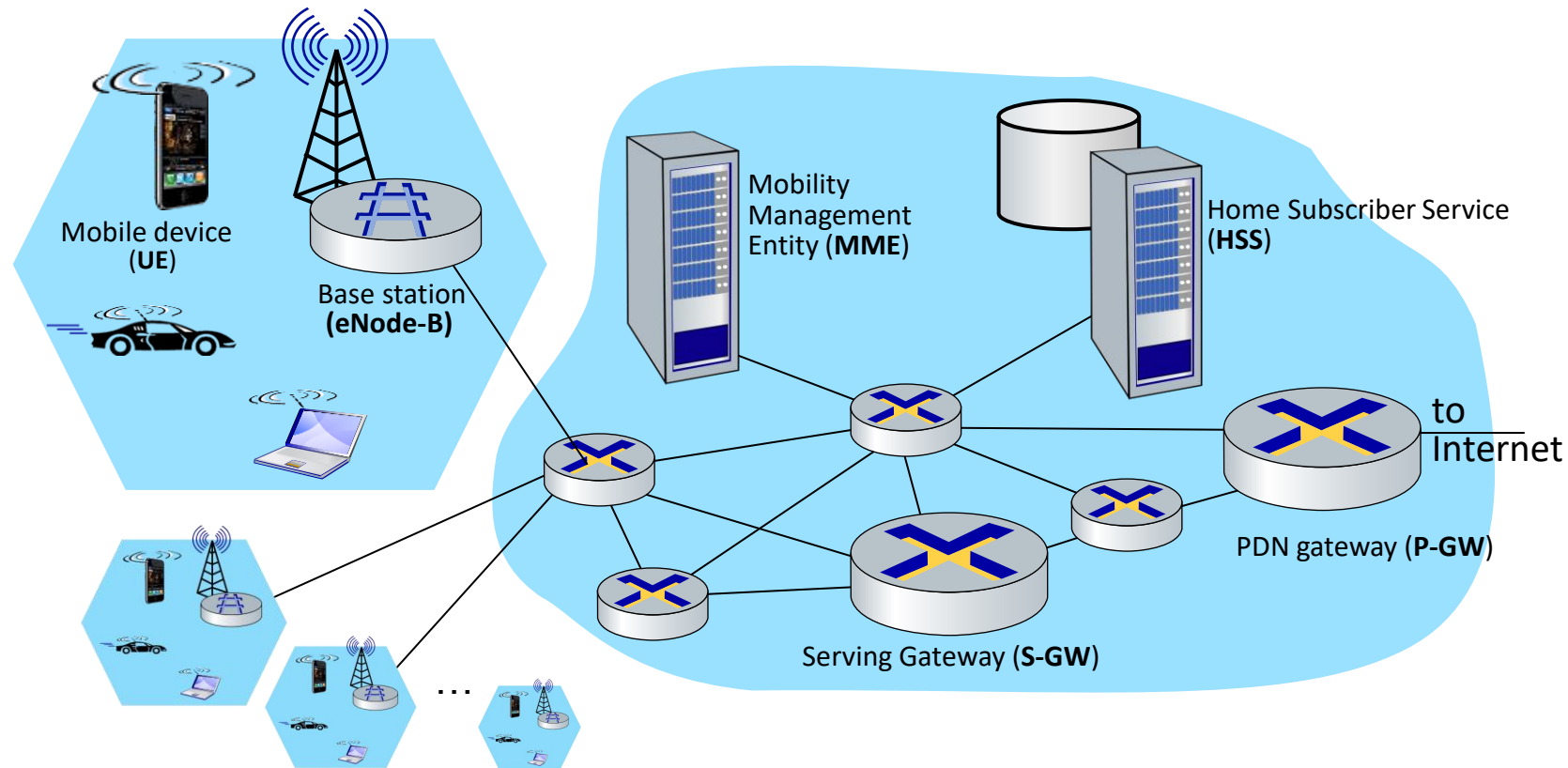


**Connect
Real-time**



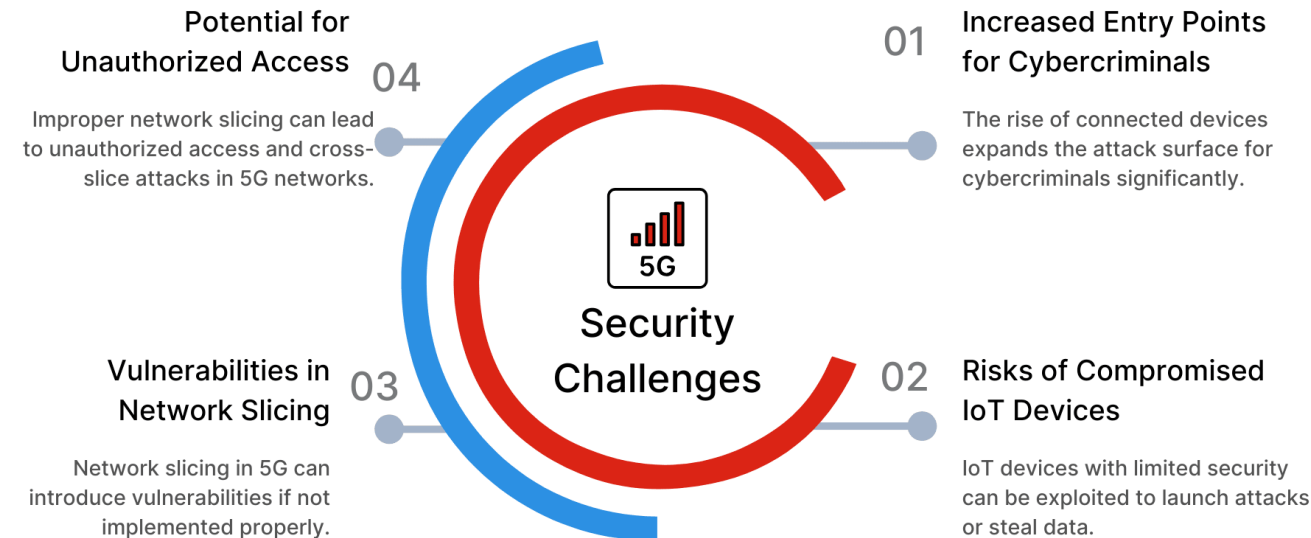
4G Network Architecture

- LTE (Long Term Evolution) uses a simpler and faster architecture than 3G

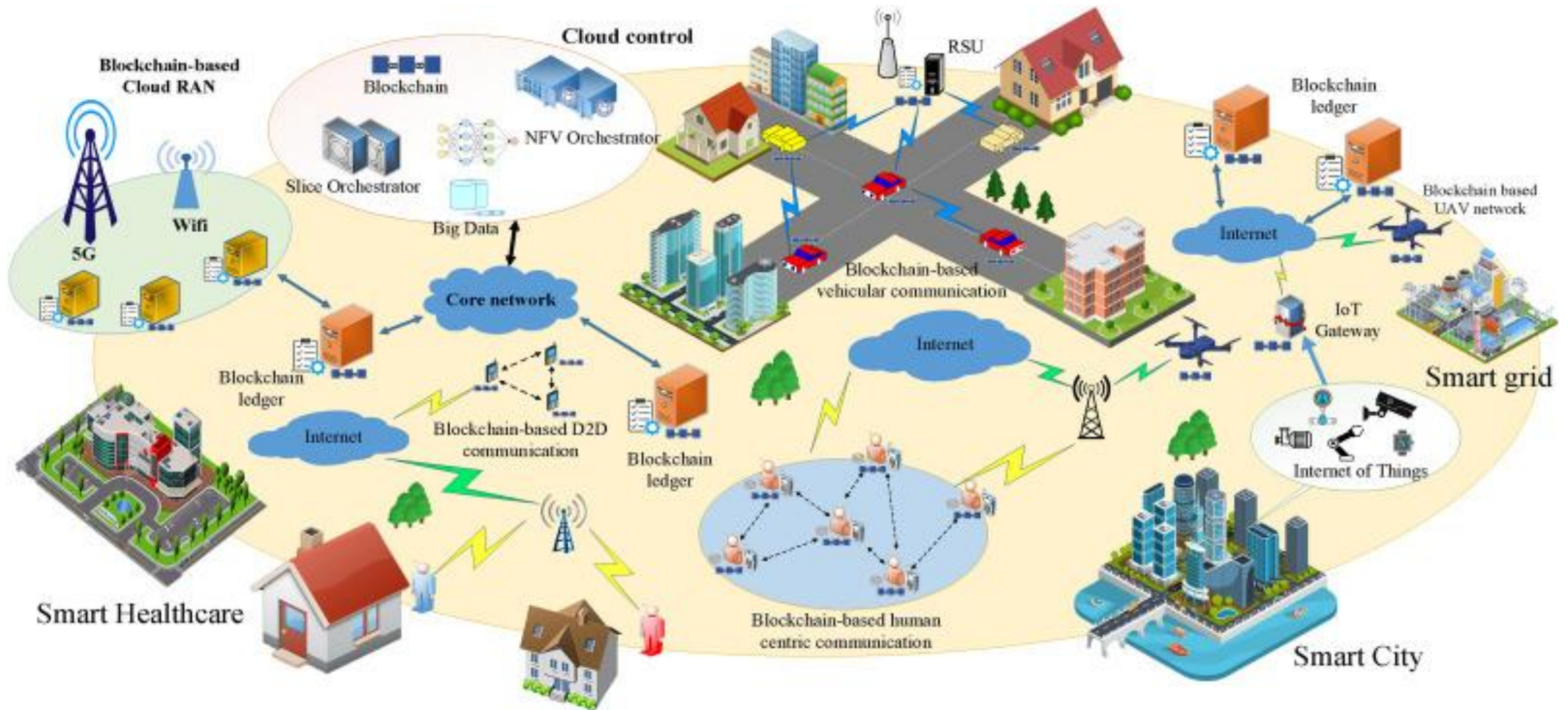


Fifth Generation (5G)

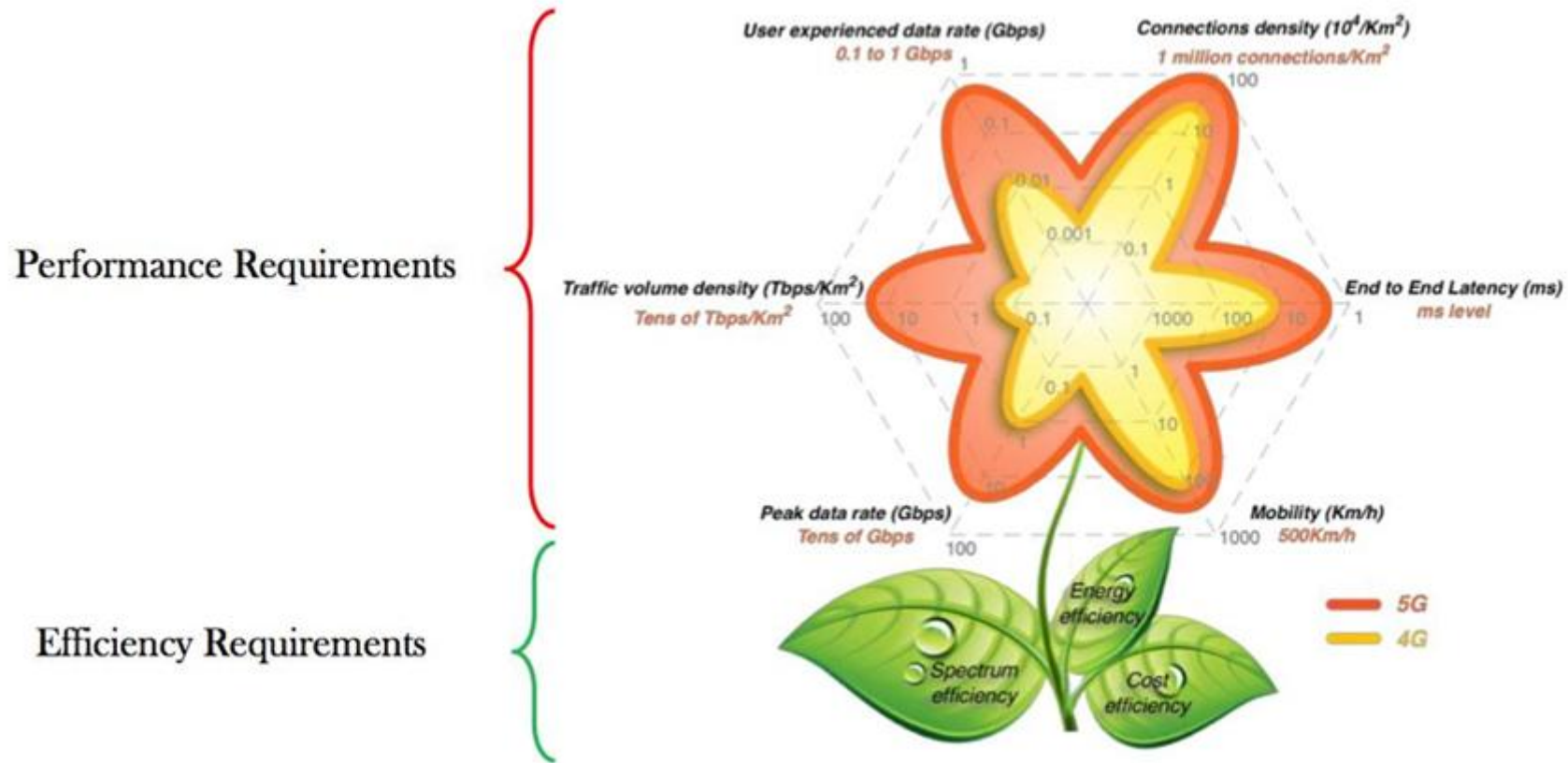
- **Enhanced Mobile Broadband Communications (eMBB)** that delivers up to Gbps speeds.
- **Massive Machine to Machine Communications (mMMC)** that supports high density (up to 1 million devices per square kilometer) and unique IoT requirements.
- **Ultra-high Reliability and Low Latency Communications (uRLLC)** enables the high reliability and very low latency that are critical for many industrial use cases.
- **Network Slicing** allows the delivery of isolated and customized “network slices” to meet specific traffic requirements and use cases. E.g. QoS, capacity, latency, and security.



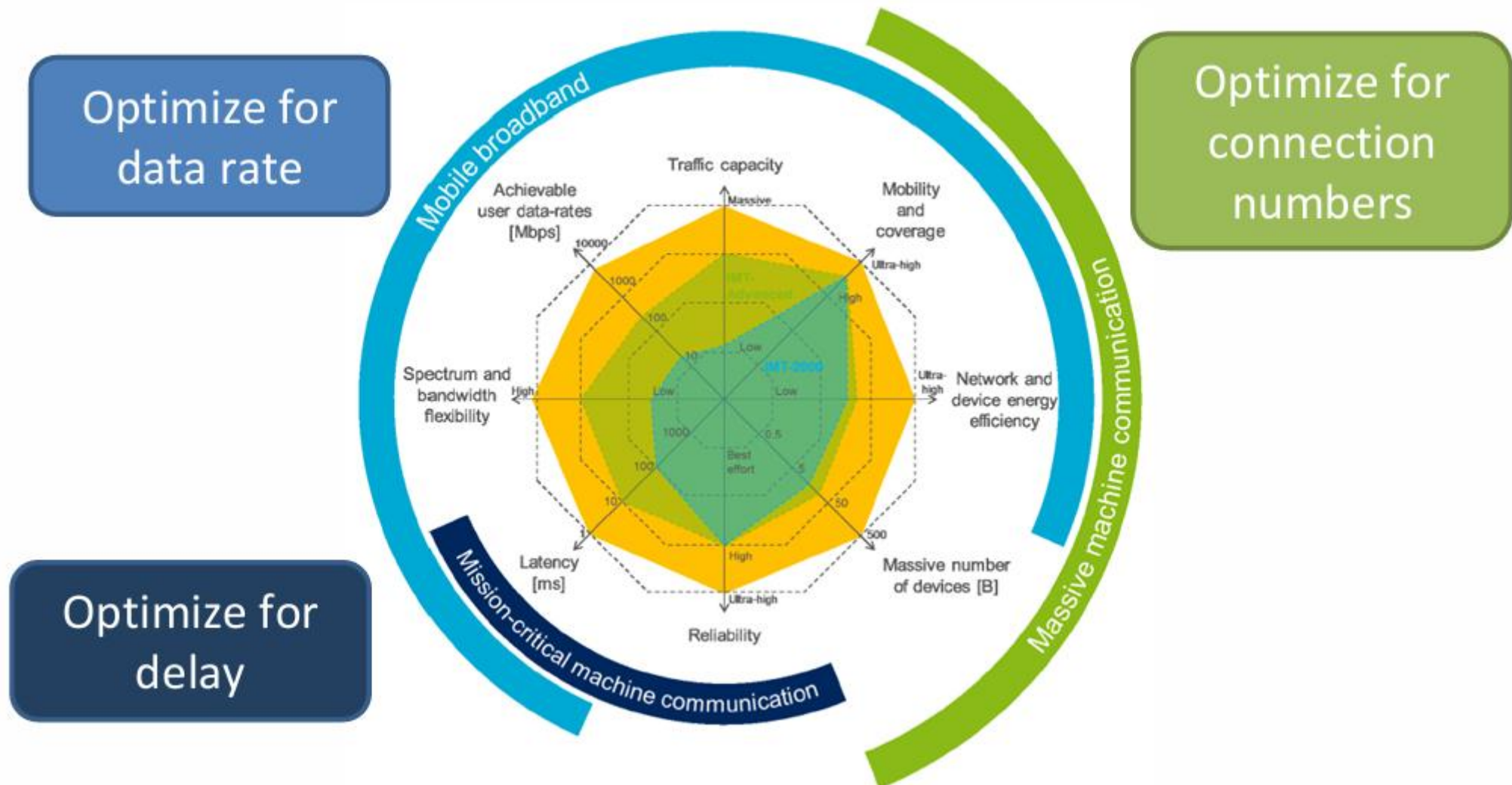
5G Network Overview



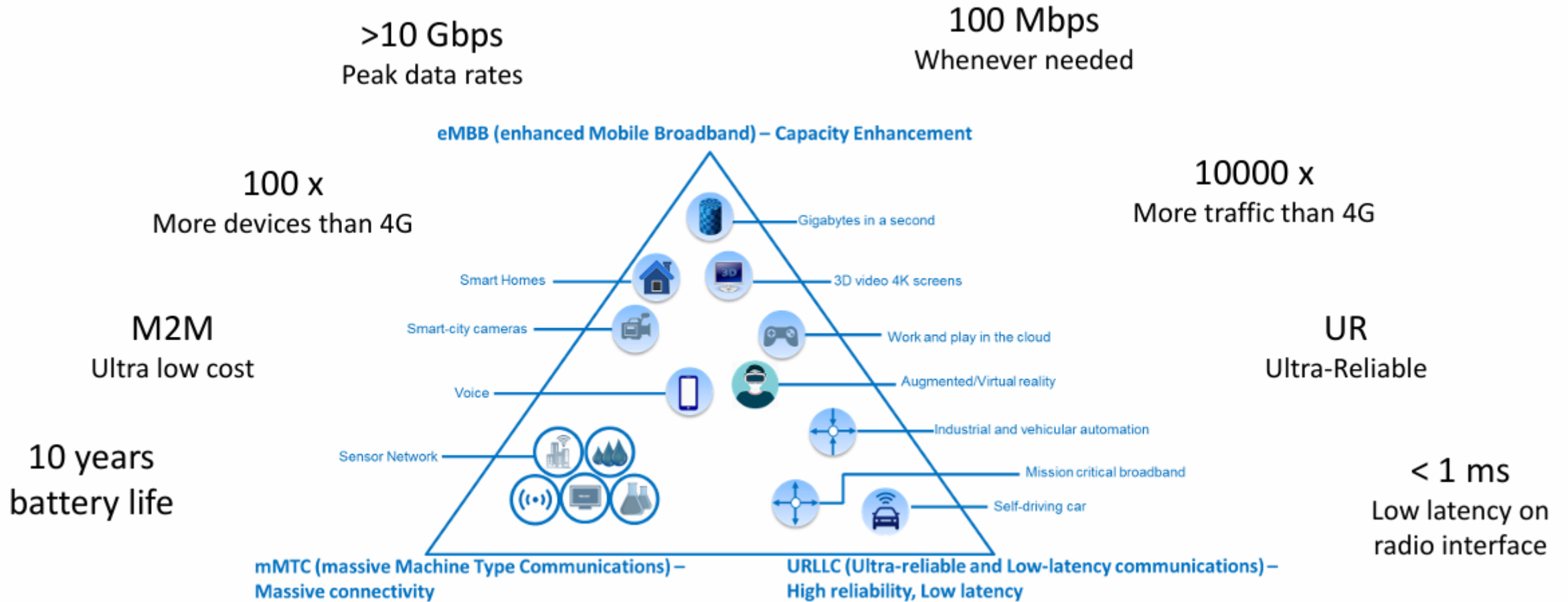
4G vs 5G



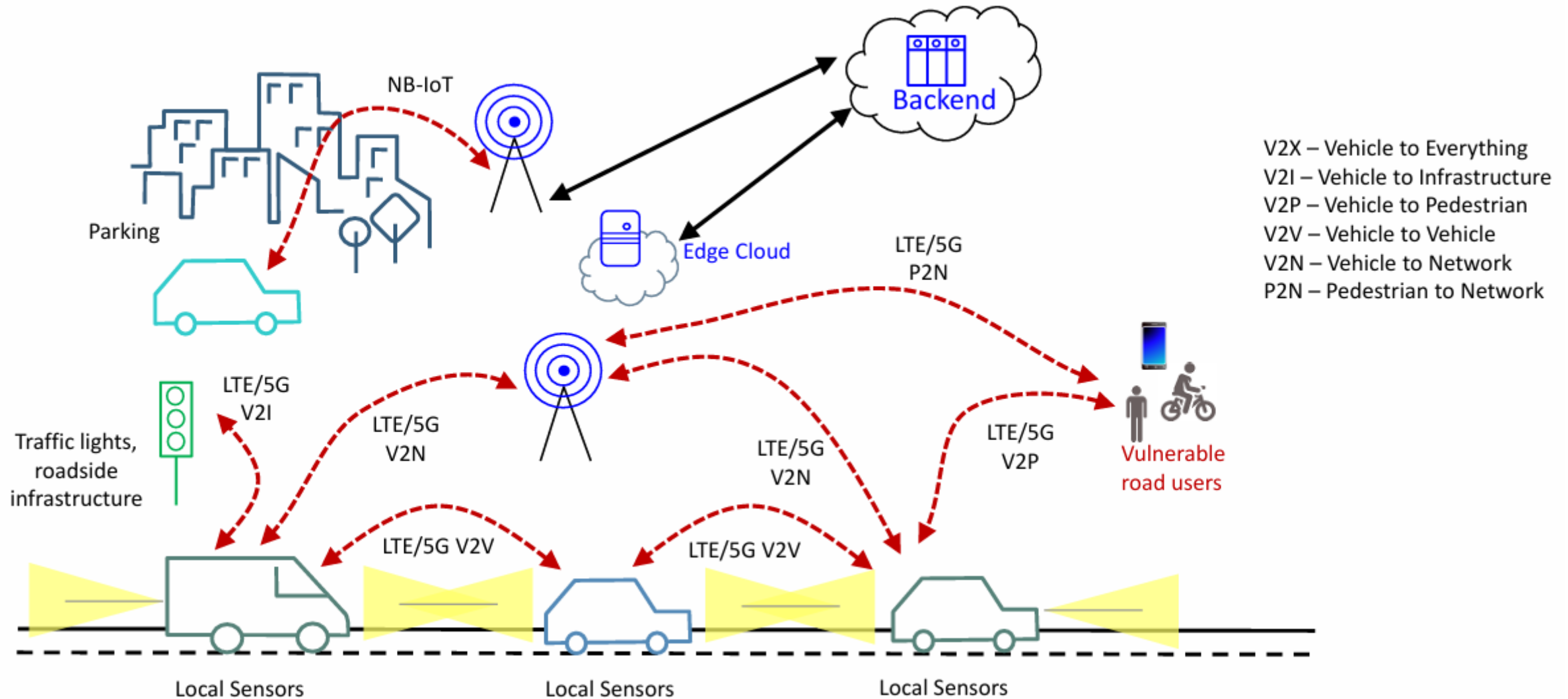
5G Requirements



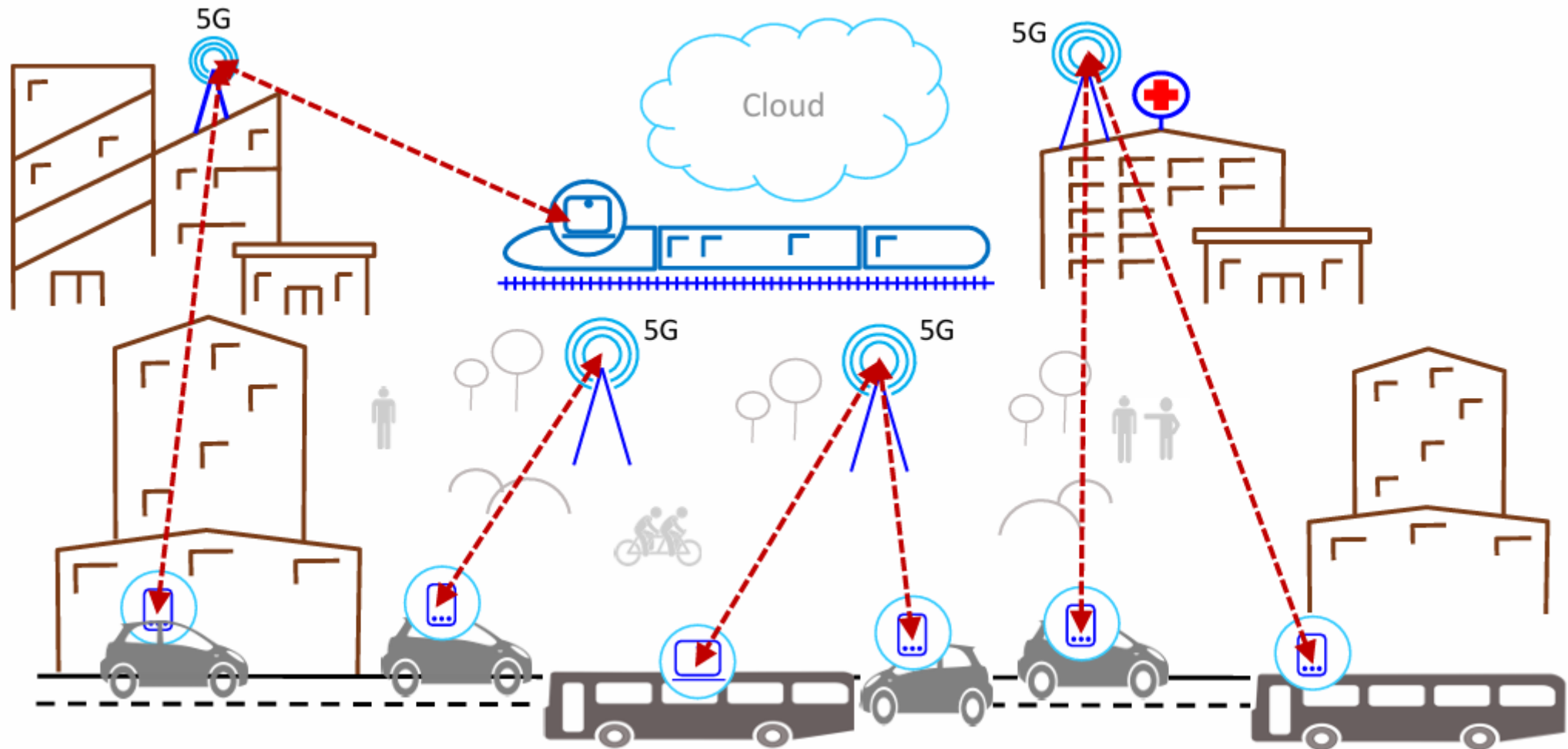
5G High Level Requirements and Wish List



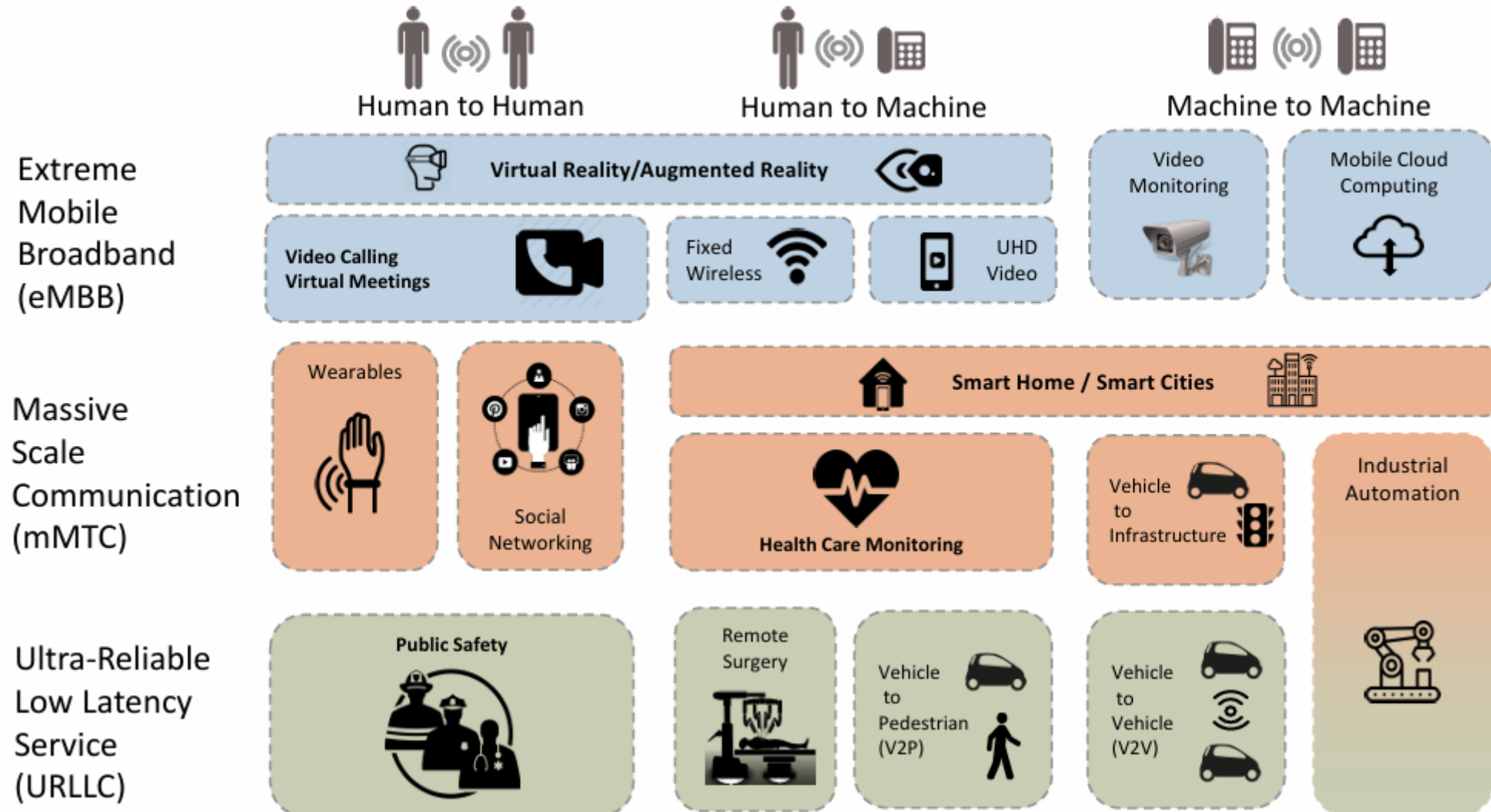
Cellular V2X Concept



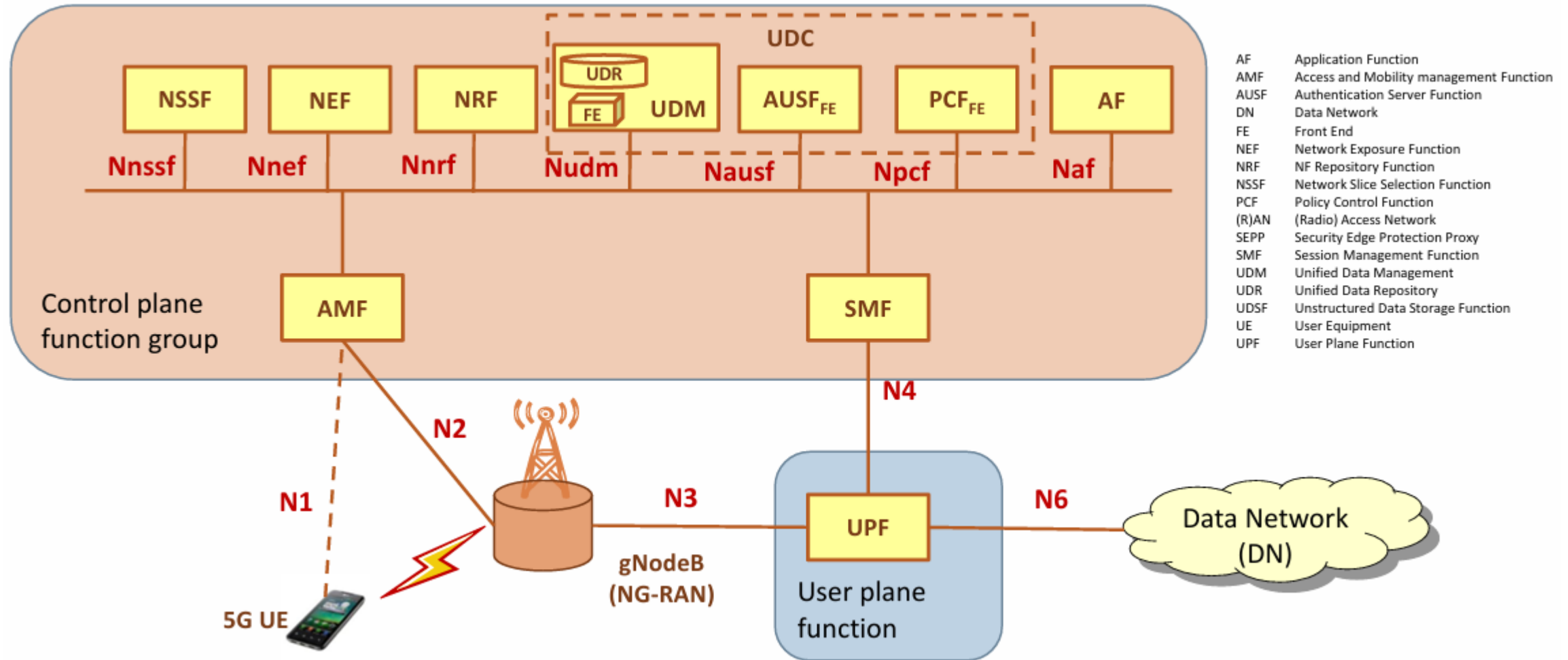
5G Connected Car: In-vehicle infotainment



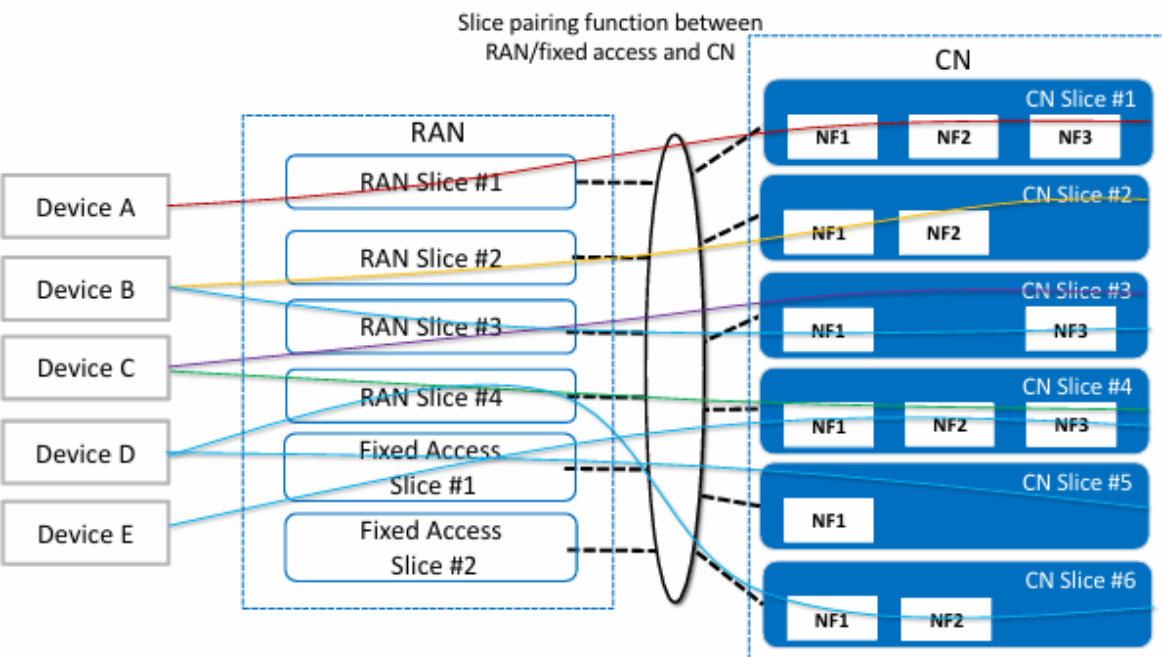
Summary of 5G Use Cases



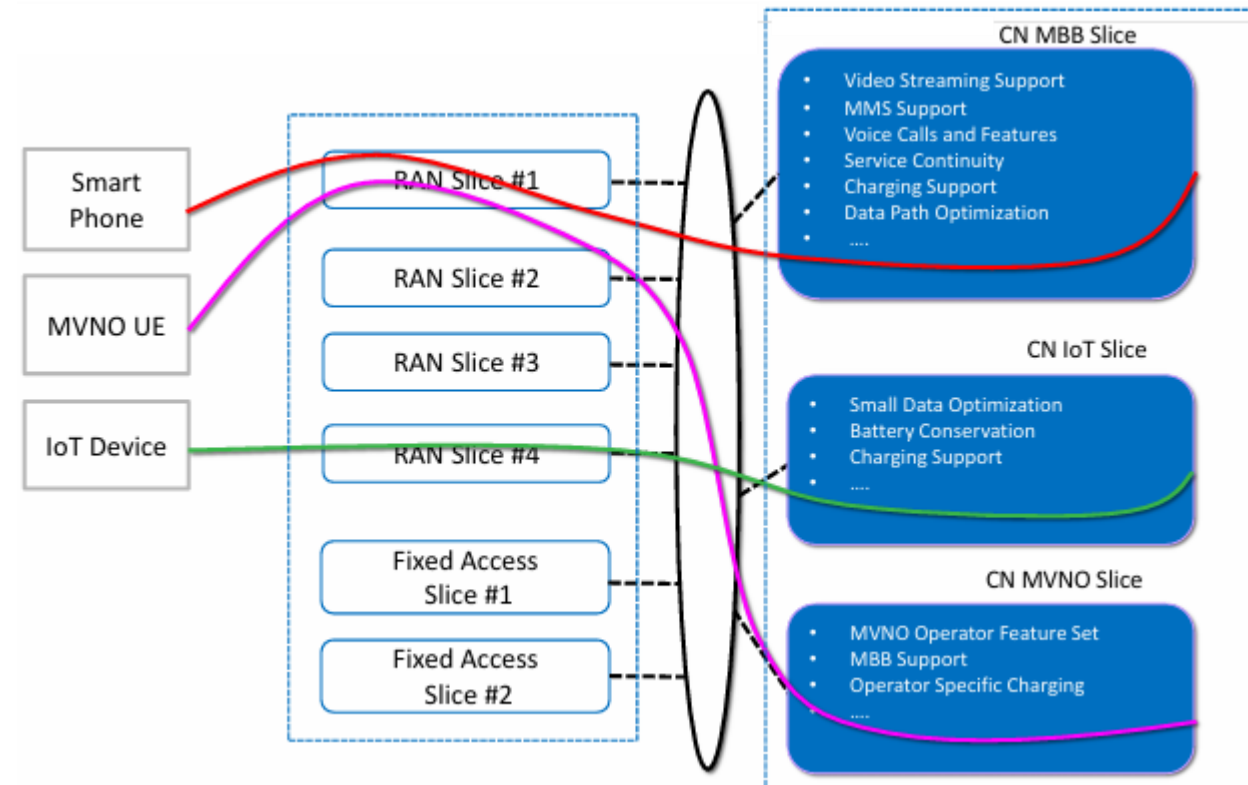
5G Network Architecture



Network Slicing



Same physical Infrastructure, different logical functions



Comparison Between different Generations

Metric	1G	2G	3G	4G / LTE	5G
Generation	First Generation	Second Generation	Third Generation	Fourth Generation	Fifth Generation
Launch Period	1980s	1990s	2000s	2010s	2020s
Technology Type	Analog (AMPS)	Digital (GSM, CDMA)	Digital Broadband (UMTS, CDMA2000)	All-IP Digital (LTE)	Advanced All-IP (5G NR)
Data Speed	2.4 kbps	64 kbps – 144 kbps	Up to 2 Mbps	100 Mbps – 1 Gbps	1 Gbps – 20 Gbps
Switching Technique	Circuit Switching	Circuit Switching	Circuit + Packet Switching	Packet Switching	Packet Switching
Core Network	PSTN-based	Digital Core	CS + PS Core	EPC	5GC (5G Core)
Internet Support	No	Limited	Yes	Full Broadband	Ultra-fast Broadband
Voice Service	Analog Voice	Digital Voice	Voice + Video Calls	VoLTE	VoNR
Latency	Very High	High	Moderate	Low (~10 ms)	Very Low (~1 ms)
Frequency Band	800 MHz	900/1800 MHz	850/1900/2100 MHz	Multiple LTE Bands	Sub-6 GHz + mmWave
Bandwidth	Very Low	Low	Medium	High	Very High
Architecture Complexity	Simple	Moderate	Complex	Simplified IP	Cloud-native
Network Type	Analog Cellular	Digital Cellular	Mobile Multimedia	Mobile Broadband	Intelligent Connectivity
Security	Very Weak	Moderate	Better Encryption	Strong Security	Advanced Security
Services Supported	Voice Calls	SMS, Voice	Internet, Video Calls	HD Streaming, VoLTE	IoT, AI, AR/VR
Handover Quality	Poor	Better	Improved	Fast	Ultra-fast
Energy Efficiency	Low	Moderate	Moderate	Better	Optimized
Device Capability	Basic Phones	Feature Phones	Smartphones	Smart Devices	AI/IoT Devices

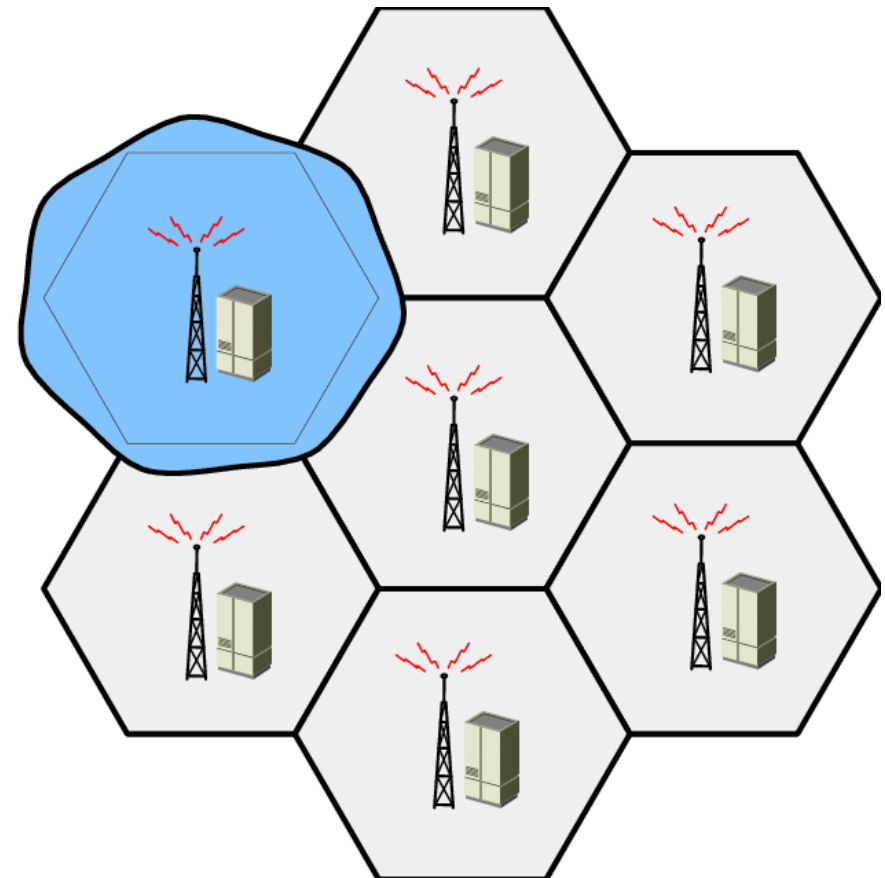
Introduction to Cellular Concept

Developed for mobile radio telephone

- To replace high power transmitter/receiver systems
- Use lower power, shorter range, more transmitters
 - Multiple low power transmitters
 - 100w or less

Area divided into cells

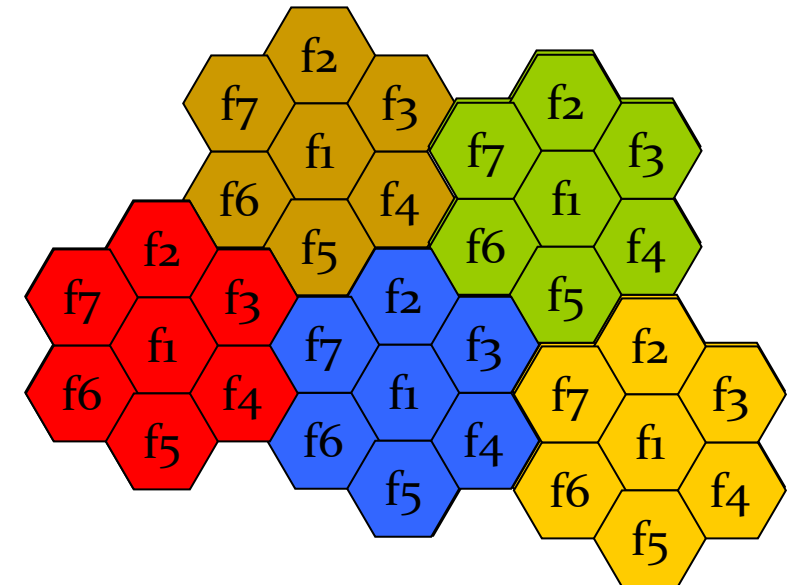
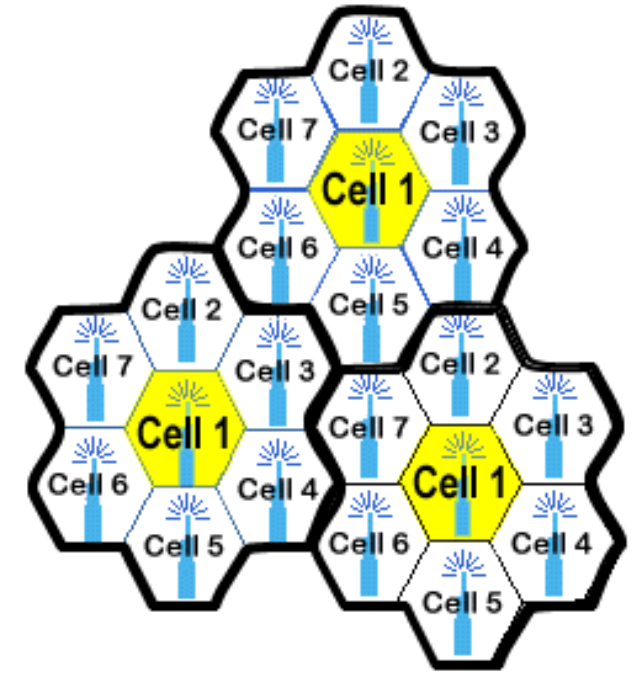
- Each with own antenna
- Each with own range of frequencies
- Served by base station
 - Transmitter, receiver, control unit
- Adjacent cells on different frequencies to avoid crosstalk
- Cell concept was proposed by Bell Labs in 1971



Cell Structure

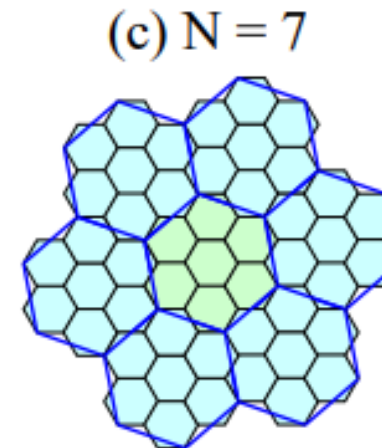
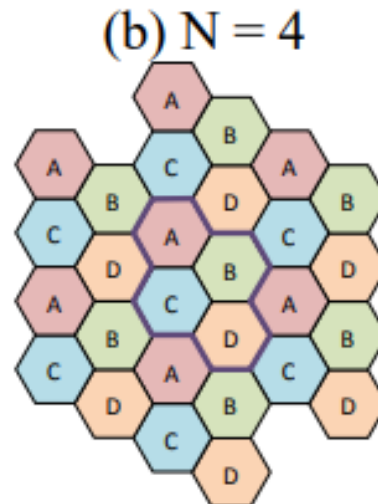
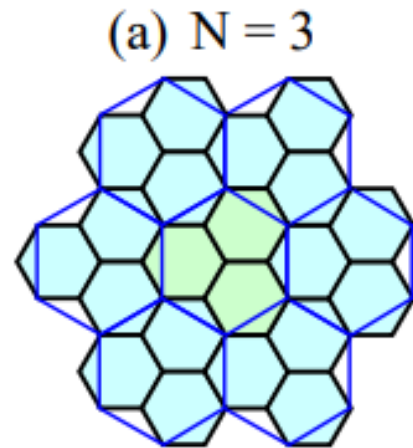
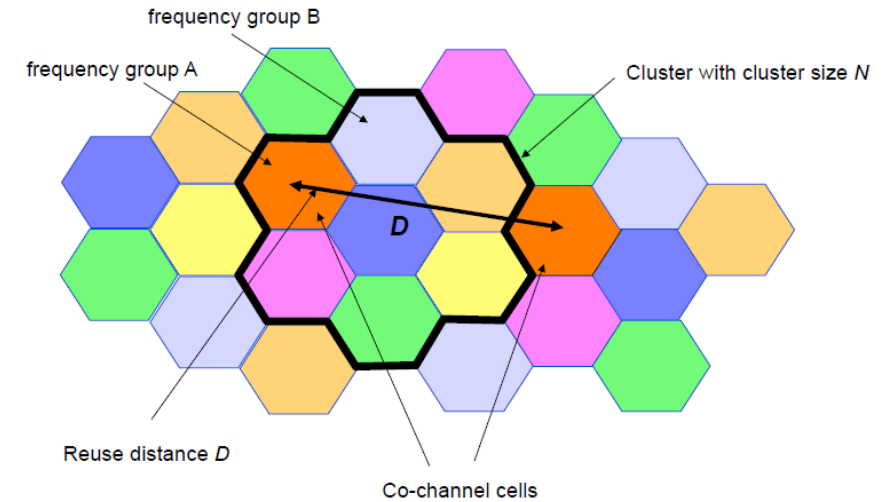
Frequency Reuse

- Each **cell** is assigned a group of **radio channels** used within a small geographic area.
 - ✓ Cells are assigned a group of channels that is completely **different** from neighbouring cells
- **Footprint** - the coverage area of cells
 - ✓ limited by a boundary so that the same group of channels can be used in cells that are far enough apart
- **Co-channel cells** - Cells with the same number have the same set of frequencies



Clusters

- The repeating regular **pattern of cells**.
- The total coverage area is divided into clusters
- There can be **no co-channel interference** within a cluster
- **Cluster size** – the number of cells in a cluster.
 - ✓ Denoted by N



Capacity

■ Let

- S = total number of available duplex radio channels for the system
- K = number of channels allocate to each cell ($k < S$)
- N = cluster size

A. If the S channels are divided among N cells into unique and disjoint channel groups with each having the same number of channels,

$$S = kN$$

B. If a cluster is replicated M times within the system, the total number of duplex channels, C , is

$$C = MS = MkN$$

C. For a fixed total coverage area A_{total} and the coverage area A_{cell} of each cell, the number of cells in the system is

$$M \times N = \frac{A_{total}}{A_{cell}}$$

$$C = \frac{A_{total}}{A_{cell}} \times \frac{S}{N}$$

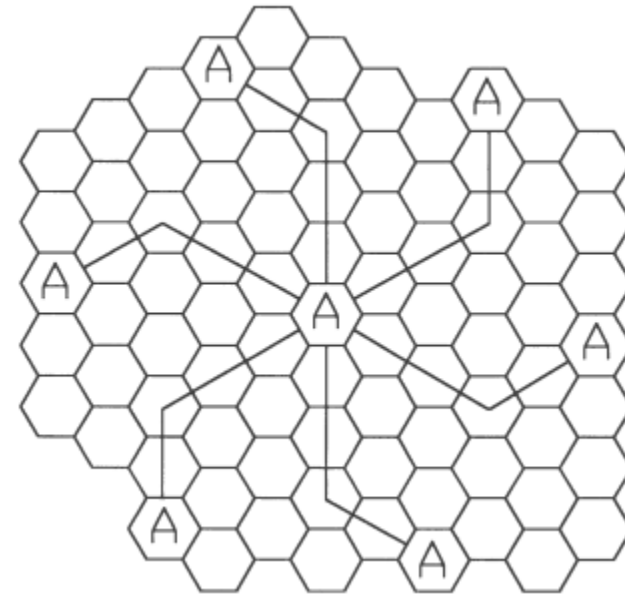
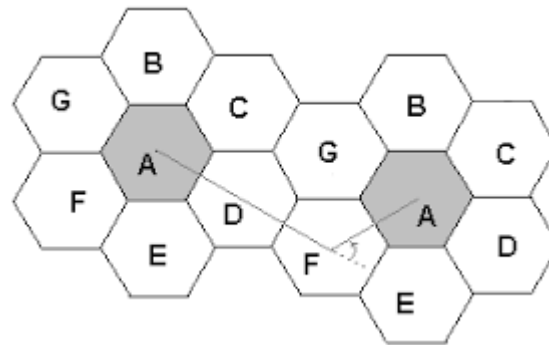
If the cluster size N is reduced while the cell size is kept constant, more clusters are required to cover a given area, and hence more capacity (a larger value of C) is achieved.

Cell Planning

- Only certain cluster sizes and cell layouts which are possible.
- The number of cells per cluster, N , can only have values which satisfy;

$$N = i^2 + ij + j^2, \quad i \geq 0, j \geq 0 \quad \text{where } i \text{ and } j \text{ are non-negative integers.}$$

- To locate the nearest co-channel neighbors of a particular cell,
 - move i cells along any chain of hexagons and then
 - turn 60 degrees counter-clockwise and move j cells



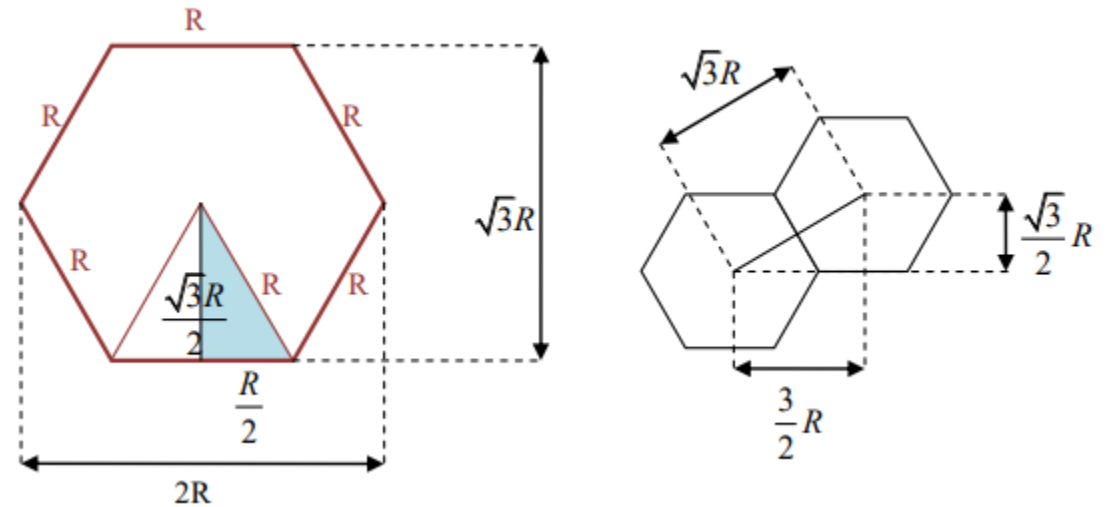
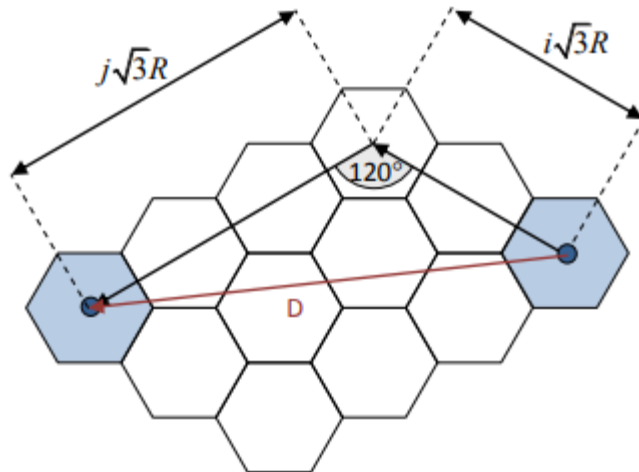
Cont...

Area of Hexagon

Let R be the radius of a Cell

$$A = 6 \times 2 \times \left(\frac{1}{2} \times \frac{\sqrt{3}}{2} R \times \frac{1}{2} R \right)$$

$$A = 12 \times \frac{\sqrt{3}}{8} R^2 \approx 2.598 R^2$$



$$C = \sqrt{a^2 + b^2 - 2ab \cos c}$$

Center-to-center distance between closest co-channels

Also called reuse distance.

$$D = \sqrt{(i\sqrt{3}R)^2 + (j\sqrt{3}R)^2 - 2(i\sqrt{3}R)(j\sqrt{3}R) \cos 120^\circ}$$

$$D = R\sqrt{3(i^2 + j^2 + ij)}$$

$$D = R\sqrt{3N}$$

Example

What would be the minimum distance between the centers of two cells with the same band of frequencies if cell radius is 1 km and the reuse factor is 12?

Solution

$$D = R\sqrt{3N}$$

$$D = 1 \text{ km} \sqrt{3 \times 12}$$

$$D = 6 \text{ km}$$

Example

- A total of 33 MHz bandwidth is allocated to a FDD cellular system with two 25 KHz simplex channels to provide full duplex voice and control channels. Compute the number of channels available per cell if the system uses
 - i. 4 cell,
 - ii. 7 cell,
 - iii. 8 cell reuse technique.

Assume 1 MHz of spectrum is allocated to control channels. Give a distribution of voice and control channels.

Solution

One duplex channel = $2 \times 25 = 50 \text{ kHz}$ of spectrum.

Hence the total available duplex channels are

$$= \frac{33 \text{ MHz}}{50 \text{ kHz}} = 660 \text{ in number}$$

Among these channels,

Control channels.

$$\frac{1 \text{ MHz}}{50 \text{ kHz}} = 20 \text{ channels}$$

(a) For $N = 4$,

$$\text{Total channels per cell} = \frac{660}{4} = 165$$

$$\text{Control channels per cell} = \frac{20}{4} = 5$$

Among these, voice channels are 160, control channels are 5 in number.

(a) For $N = 7$,

$$\text{Total channels per cell} = \frac{660}{7} = 94$$

$$\text{Control channels per cell} = \frac{20}{7} = 2$$

Here, 6 cells can use 3 control channels and the other one channel can use 2 control channels each. On the other hand, 5 cells can use 92 voice channels and the rest two can use 90 voice channels each.

$$\begin{aligned} \text{control channels} &= (6 \times 3) + (1 \times 2) = 20 \\ \text{voice channels} &= (5 \times 92) + (2 \times 90) = 640 \end{aligned}$$

This is one solution, there might exist other solutions too.

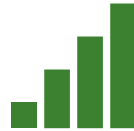
(a) For $N = 8$,

The option $N = 8$ is not a valid option since it cannot satisfy equation $N = i^2 + ij + j^2$ by two integers i and j .

Co-Channel Interference



Interference is the major limiting factor in the performance of cellular radio systems. channel interference.



Sources of interference include

another mobile in the same cell
a call in progress in a neighboring cell,
other base stations operating in the same frequency band, or
any non-cellular system which inadvertently leaks energy into the cellular frequency band.



Effects of interference:

Interference on voice channels causes cross talk
On control channels, interference leads to missed and blocked calls due to errors in the digital signaling.



The two major types of system-generated cellular interference are

co-channel interference
adjacent channel interference.

Co-channel Reuse Ratio

For a hexagonal geometry, the parameter Q called the frequency reuse ratio is given as

$$Q = \frac{D}{R} = \sqrt{3N}$$

If D/R ratio is increased, then the effective distance between the co-channel cells will increase and interference will decrease.

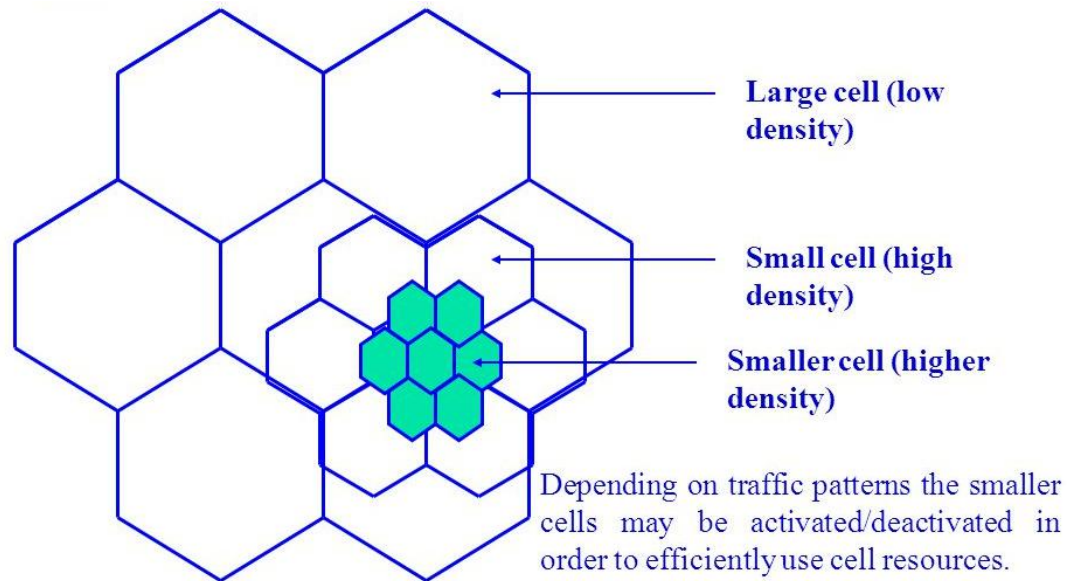
- Small Q means small value of cluster size N and increase in cellular capacity. But large Q leads to decrease in system capacity but increase in transmission quality.

Improving Coverage and Capacity

- Cellular design techniques used to provide more channels per unit coverage area include;
 - ✓ cell splitting
 - ✓ sectoring

Cell Splitting

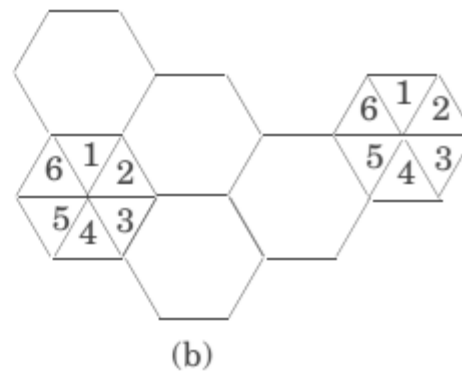
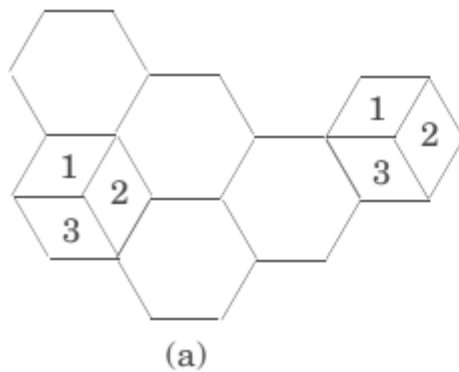
- It is the process of subdividing a congested cell into smaller cells (called microcells)
- Each microcell has its own base station and a corresponding reduction in antenna height and transmitter power.



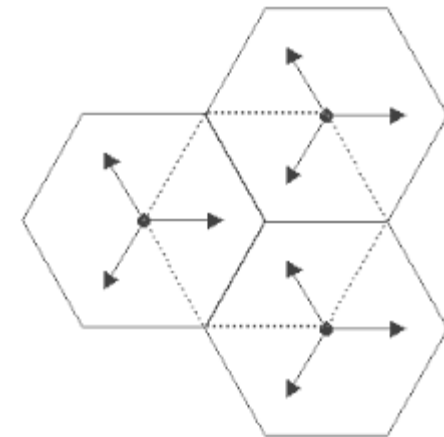
- Cell splitting increases the capacity of a cellular system since it increases the number of times that channels are reused.

Sectoring

- Sectoring is a method of decreasing the co-channel interference and enhancing system performance by using a directional antenna.
- Replace a single omnidirectional antenna at the base station by several directional antennas, each radiating within a specified sector.



(a) 120 deg sectoring; (b) 60 deg sectoring.



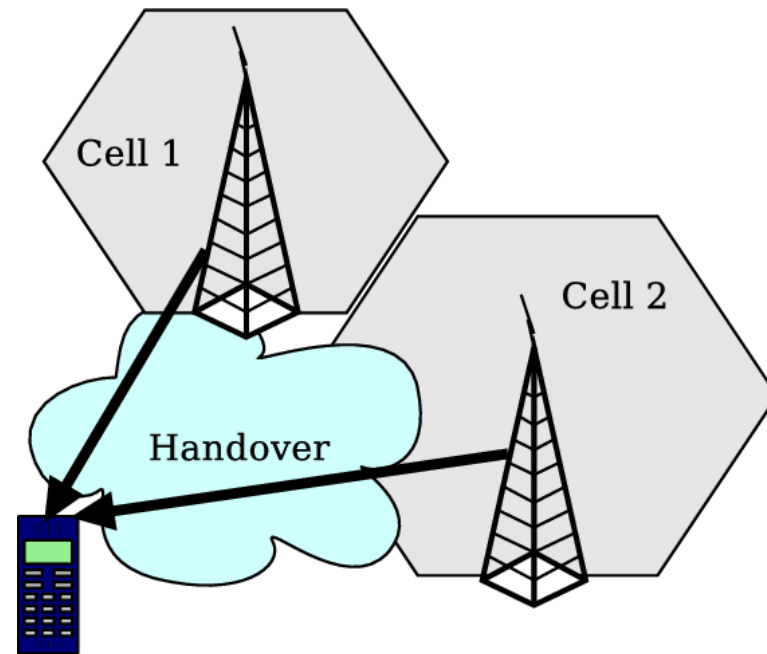
Handoffs

- A handoff is the process of transferring an active call or data session from one cell in a cellular network to another or from one channel in a cell to another.

- ✓ A well-implemented handoff is important for delivering uninterrupted service to a caller or data session user.

- **Factors Influencing Handoffs**

- i. Transmitted power - handoff threshold
- ii. Received power - Line of Sight (LoS) factor
- iii. Area and shape of the cell
- iv. Mobility of users:



Handoff Cont...

Types of Handoff

- **Soft Handoff**
 - A communication is handed off to another cell without interrupting the communications serving radio port.
 - A user is shifted from one BS to the other without actually changing the channel.
- **Hard Handoff**
 - An existing connection must be broken before the new one is established.
 - Hard handoff allocate different frequency of user.

Handoff Process Steps

- **Initiation** – Either the mobile terminal or the network identifies the need for a handoff and alerts the necessary network elements.
- **Resource Reservation** – the appropriate network elements reserve the resources necessary to support the handoff.
- **Execution** – the actual handoff connection of the network resource takes place.
- **Completion** – any unneeded network resources are freed, and access signals are exchanged following a successful handoff.